Building and evaluation of a PBPK model for Paroxetine in CYP2D6 phenotype and activity-score groups

Version	evaluation-OSP12.2
based on *Model Snapshot* and *Evaluation Plan*	https://github.com/Open-Systems-Pharmacology/Paroxetine-Model/releases/tag/vevaluation
OSP Version	12.2
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This evaluation report and the corresponding PK-Sim project file are filed at:

https://github.com/Open-Systems-Pharmacology/OSP-PBPK-Model-Library/

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1 Introduction

Paroxetine is a selective serotonin reuptake inhibitor used for the treatment of depression and other psychiatric indications. Paroxetine is administered orally and undergoes nonlinear metabolism, mainly via CYP2D6, with additional CYP3A4 metabolism and CYP2D6-independent hepatic clearance represented in the model.

The whole-body PBPK model published by Rüdesheim 2022 is intended to describe paroxetine plasma concentration-time profiles across CYP2D6 phenotype and activity-score groups and to support CYP2D6 drug-gene interaction simulations.

This whole-body PBPK model of paroxetine has been developed using published pharmacokinetic clinical data by Lund 1982, Belle 2002, Massaroti 2005, Mürdter 2016, Sindrup 1992, and Yoon 2000. The clinical data included intravenous infusion, immediate-release oral dosing, tablet dosing, controlled-release oral dosing, and repeated dosing across a dose range of 20 to 70 mg paroxetine. The model was then evaluated with clinical studies including Calvo 2004, Chen 2015, McClelland 1984, Schoedel 2012, Segura 2005, van der Lee 2007, Yasui-Furukori 2006, Yasui-Furukori 2007, Mürdter 2016, and Yoon 2000.

The presented model includes the following features:

- CYP2D6-mediated metabolism with activity-score specific k_{cat} values,
- CYP3A4-mediated metabolism,
- mechanism-based inactivation of CYP2D6 and CYP3A4 by paroxetine,
- unspecific hepatic clearance,
- renal filtration with a GFR fraction of 1,
- intravenous infusion, immediate-release oral, tablet, and controlled-release oral applications.

2 Methods

2.1 Modeling strategy

The general concept of building a PBPK model has previously been described by Kuepfer et al. (Kuepfer 2016). Relevant information on anthropometric (height, weight) and physiological parameters (e.g. blood flows, organ volumes, binding protein concentrations, hematocrit, cardiac output) in adults was gathered from the literature and has been previously published (Willmann 2007). The information was incorporated into PK-Sim® and was used as default values for the simulations in adults.

The applied activity and variability of plasma proteins and active processes that are integrated into PK-Sim® are described in the publicly available PK-Sim® Ontogeny Database Version 7.3 (PK-Sim Ontogeny Database Version 7.3) or otherwise referenced for the specific process.

The paroxetine base model was developed sequentially. Model-building data from intravenous paroxetine administration in Lund 1982 supported the selection of partition-coefficient and cellular-permeability calculation methods. Oral single-dose and multiple-dose data from Lund 1982, Belle 2002, Massaroti 2005, Sindrup 1992, Mürdter 2016, and Yoon 2000 informed oral absorption, CYP2D6-independent metabolism, and CYP2D6-mediated metabolism.

Oral solution and tablet applications were used for immediate-release dosing, while controlled-release oral applications were represented with Weibull dissolution parameters based on controlled-release formulation data. The model contains CYP2D6 and CYP3A4 as active metabolic processes, an unspecific hepatic clearance process, renal filtration, and mechanism-based inhibition processes for CYP2D6 and CYP3A4.

For the CYP2D6 drug-gene interaction model, CYP2D6-dependent clearance was represented with Michaelis-Menten kinetics. CYP2D6 k_{cat} values were optimized for activity score 2 and set to zero for poor metabolizers, while other activity-score k_{cat} values were calculated from a polynomial regression derived from metoprolol and dextromethorphan models. CYP2D6 K_m and CYP2D6 reference concentrations were kept constant over the modeled activity-score range.

Details about the input data are provided in Section 2.2. Details about the structural model and assumptions are provided in Section 2.3.

2.2 Data used

In vitro and physicochemical data

The table below summarizes the drug-dependent inputs documented for the paroxetine model.

Parameter	Unit	Value	Source	Description
MW	g/mol	329.37	Zhong 2013	Molecular weight.
pK_a,base	-	9.9	Agrawal 2013	pKa of a basic ionization site.
Solubility	mg/mL	7.31	Khatavkar 2016	Aqueous solubility at pH 4.5.
logP	-	3.95	Agrawal 2013	Octanol-water partition coefficient of the neutral species.
f_u	%	4.873[d]	Optimized; Kaye 1989	Fraction unbound in plasma.
$K_{m,CYP3A4}$	µmol/L	4.7	Jornil 2010	Michaelis constant for CYP3A4-mediated metabolism.
$k_{cat,CYP3A4}$	1/min	1.01	Optimized; Jornil 2010	Catalytic rate constant for CYP3A4-mediated metabolism.
$K_{m,CYP2D6}$	µmol/L	0.028	Jornil 2010	Activity-score-invariant Michaelis constant for CYP2D6-mediated metabolism.
$k_{cat,CYP2D6}$, EM	1/min	1.3693020782	Optimized; Jornil 2010	Catalytic rate constant for the extensive-metabolizer pathway.
$k_{cat,CYP2D6}$, PM	1/min	0	Assumed; Jornil 2010	Catalytic rate constant for the poor-metabolizer pathway.
CL_{hep}	1/min	2.2538632474[c]	Optimized; Jornil 2010	Unspecific hepatic clearance.
$K_{I,CYP2D6}$	µmol/L	0.16738	Optimized; Venkatakrishnan 2005	Inactivator concentration at half-maximal inactivation rate of CYP2D6.
$k_{inact,CYP2D6}$	1/min	0.17	Venkatakrishnan 2005	Maximum CYP2D6 inactivation rate constant.
$K_{I,CYP3A4}$	µmol/L	4.48	Bertelsen 2003	Inactivator concentration at half-maximal inactivation rate of CYP3A4.
$k_{inact,CYP3A4}$	1/min	0.011	Bertelsen 2003	Maximum CYP3A4 inactivation rate constant.
GFR fraction	-	1	Assumed	Fraction used to scale passive glomerular filtration.
Weibull shape	-	7.17	Optimized	Shape parameter for controlled-release dissolution.
t_{50}	min	276.35	Optimized; Chen 2015; Khatavkar 2013	Time to dissolve 50% of the controlled-release dose.

Parameter	Unit	Value	Source	Description
Partition coefficients	-	Rodgers and Rowland	Calculated; Rodgers 2007	Tissue-to-plasma partition coefficients calculated with the Rodgers and Rowland method.
Cellular permeabilities	-	PK-Sim Standard	Calculated (Kawai 1994)	Cellular permeabilities calculated with the PK-Sim Standard method.
P_{int}	cm/min	3.93×10^{-5}	Optimized	Specific transcellular intestinal permeability.

Table 1: Drug-dependent physicochemical, distribution, metabolism, elimination, inhibition, and formulation parameters used in the final paroxetine model.

[c] The optimized model value is retained. The supplement to Rüdesheim 2022 reports 1.37 1/min.

[d] The optimized model value is retained. The supplement to Rüdesheim 2022 reports the literature value 5.00% (Kaye 1989).

The CYP2D6 activity-score-specific catalytic rate constants implemented in the model are listed below.

CYP2D6 AS	k_{cat} [1/min]	Origin
0	0.00	Assumed
0.5	0.66	Calculated
0.75	1.08	Calculated
1	1.56	Calculated
1.25	2.11	Calculated
1.5	2.71	Calculated
2	4.09	Optimized
3	7.58	Calculated

Table 1a: CYP2D6 activity-score-specific k_{cat} values for paroxetine metabolism from Table 2 of Rüdesheim 2022. The AS = 2 value was optimized, AS = 0 was assumed to have no CYP2D6 activity, and the other implemented values were calculated from the relative activity scale. AS: activity score.

Clinical data

The evaluation uses 33 plasma concentration-time profiles from adults (Table 2). Fourteen profiles were used for model building and 19 profiles were used for model verification. The assignments follow the training and test classifications in the published model supplement. For interaction studies, this report includes only paroxetine-only or control-arm pharmacokinetic data.

Source	Dose [mg] / schedule*	Age [years]	Weight [kg]	Sex	N	Form.	CYP2D6 characterization
Belle 2002+	20 q.d., 12 days	38 (20–49)	NR	23% female	22	Tablet	EM
Calvo 2004	20 q.d., 15 doses	26	64	64% female	25	NR	EM
Chen 2015	25	26 (19–45)	64	25% female	4	Tablet CR	AS = 0.5 (IM)
Chen 2015	25	26 (19–45)	61	45% female	11	Tablet CR	AS = 1 (IM)
Chen 2015	25	22 (19–45)	58	60% female	5	Tablet CR	AS = 1.5 (NM)
Chen 2015	25	28 (19–45)	61	25% female	4	Tablet CR	AS = 2 (NM)
Lund 1982+	23, intravenous infusion	29	72	Male	1	Solution	EM
Lund 1982+	28, intravenous infusion	28	75	Male	1	Solution	EM
Lund 1982+	28, intravenous infusion	24	66	Male	1	Solution	EM
Lund 1982+	28, intravenous infusion	26	88	Male	1	Solution	EM
Lund 1982+	45	28	75	Male	1	NR	EM
Lund 1982+	45	24	66	Male	1	NR	EM
Lund 1982+	45	26	88	Male	1	NR	EM
Lund 1982+	45	29	72	Male	1	NR	EM
Massaroti 2005+	20	28 (18–42)	72 (57–87)	Male	28	Tablet	EM
McClelland 1984	70	31 (22–44)	NR	Male	5	NR	EM
Mürdter 2016	40 q.d., 3 days	25 (22–26)	62 (50–70)	Female	3	Tablet	AS = 0 (PM)

Source	Dose [mg] / schedule*	Age [years]	Weight [kg]	Sex	N	Form.	CYP2D6 characterization
Mürdter 2016	40 q.d., 3 days	24 (21–30)	59 (56–64)	Female	4	Tablet	AS = 0.5 (IM)
Mürdter 2016	40 q.d., 3 days	25	68	Female	1	Tablet	AS = 0.75 (IM)
Mürdter 2016	40 q.d., 3 days	26 (23–28)	67 (64–74)	Female	2	Tablet	AS = 1 (IM)
Mürdter 2016[+]	40 q.d., 3 days	32 (26–43)	57 (48–64)	Female	3	Tablet	AS = 2 (NM)
Mürdter 2016	40 q.d., 3 days	26 (22–28)	62 (54–73)	Female	3	Tablet	AS = 3 (UM)
Schoedel 2012	20 q.d., 12 days	34 (19–55)	75	14% female	14	NR	EM
Segura 2005	20 q.d., 2 days	23	65	Male	7	Tablet	EM
Sindrup 1992[+]	30 q.d., 14 doses	24 (20–30)	73 (65–81)	Male	9	Tablet	EM[¶]
Sindrup 1992[+]	30 q.d., 14 doses	27 (23–39)	82 (68–95)	Male	8	Tablet	PM
van der Lee 2007	20 q.d., 10 days	44 (18–64)	69 (51–89)	69% female	26	Tablet	EM
Yasui-Furukori 2006	20	25 (20–35)	58 (46–75)	25% female	12	NR	AS = 1.25 (NM)
Yasui-Furukori 2007	20	24 (21–35)	57 (45–67)	23% female	13	NR	EM[§]
Yoon 2000	40	21	58	Female	1	Tablet	AS = 0 (PM)
Yoon 2000	40	22	68	Male	3	Tablet	AS = 0.5 (IM)
Yoon 2000	40	22	67	Male	6	Tablet	AS = 1.25 (NM)
Yoon 2000[+]	40	23	59	17% female	6	Tablet	AS = 2 (NM)

Table 2: Clinical paroxetine concentration-time profiles used for model building and verification. *: Single oral dose unless otherwise specified; AS: activity score; CR: controlled release; EM: extensive metabolizer; IM: intermediate metabolizer; NM: normal metabolizer; NR: not reported; PM: poor metabolizer; PT: predicted phenotype; q.d.: once daily; UM: ultrarapid metabolizer; [+]: data used for

model building. Parenthetical PTs for AS-coded rows use the current CPIC CYP2D6 activity score-to-phenotype mapping (Moore 2026). EM is the model default when study-specific CYP2D6 information is not available. §: CYP2D6 status not reported. AS = 1.25 was assumed because the study population is similar to Yasui-Furukori 2006. The simulation uses the AS = 1.25 k_{cat} (Table 1a). ¶: phenotyped EM group. AS = 2 was assumed because of the low sparteine metabolic ratio. The simulation uses the AS = 2 k_{cat} (Table 1a).

2.3 Model parameters and assumptions

Absorption

The model includes intravenous infusion, immediate-release oral solution, oral tablet, and controlled-release oral applications. Intravenous simulations do not require an absorption process. Oral-solution and tablet simulations are represented with dissolved formulation input, while controlled-release simulations use a Weibull tablet formulation.

The value for `Specific intestinal permeability` and the solubility at pH 4.5 are summarized in Section 2.2.1. The controlled-release formulation is described by the `CR Weibull shape` and `CR Weibull time` parameters.

Distribution

Paroxetine is highly bound to plasma proteins. The literature starting value for `Fraction unbound (plasma, reference value)` was 5.00% (Kaye 1989). The value was optimized to 4.873% in the PBPK model.

An important parameter influencing the resulting volume of distribution is lipophilicity. A `Lipophilicity` value of 3.95 was used in the model as summarized in Section 2.2.1.

Partition coefficients were calculated with the Rodgers and Rowland method. Cellular permeabilities were calculated with the PK-Sim Standard method.

Metabolism and elimination

Two metabolic pathways and one residual hepatic clearance process were implemented into the model.

- CYP2D6

CYP2D6 metabolism is represented by Michaelis-Menten kinetics. Two CYP2D6 parameter sets are used. Study arms with an EM phenotype, reported or model default, use the EM k_{cat} (Table 1). Study arms with a reported activity score use separate activity-score CYP2D6 processes (Table 1a). The EM k_{cat} and the AS = 2 k_{cat} were optimized, the AS = 0 value was set to zero, and the other activity-score values were calculated from the regression described by Rüdesheim 2022. The EM k_{cat} is 35% lower than the AS = 1.25 k_{cat}. Rüdesheim 2022 attributes this difference to the often arbitrary definition of the EM phenotype. Exceptions: the Yasui-Furukori 2007 arm has no reported CYP2D6 data and uses the AS = 1.25 k_{cat}. The phenotyped EM arm of Sindrup 1992 uses the AS = 2 k_{cat}.

- CYP3A4

CYP3A4 metabolism is represented by Michaelis-Menten kinetics using the K_m and k_{cat} values summarized in Section 2.2.1.

- Unspecific hepatic clearance

The model contains optimized unspecific hepatic clearance as summarized in Section 2.2.1.

Additionally, renal filtration is represented with a `GFR fraction` of 1. Mechanism-based inhibition processes for CYP2D6 and CYP3A4 are included using the K_i and k_{inact} values summarized in Section 2.2.1.

3 Results and Discussion

The PBPK model for paroxetine was developed and verified with clinical pharmacokinetic data after intravenous and oral administration. The evaluation covers immediate-release and controlled-release applications, single-dose and repeated-dose regimens, and a dose range of 20 to 70 mg. The clinical dataset also includes CYP2D6 poor-metabolizer, extensive-metabolizer, and activity-score stratified groups, which are central to the intended use of the model.

The model was built with data supporting distribution, oral absorption, CYP2D6-dependent clearance, CYP2D6-independent clearance, and controlled-release dissolution. Intravenous data from Lund 1982 supported systemic disposition independent of oral absorption. Oral data from Belle 2002, Massaroti 2005, Mürdter 2016, Sindrup 1992, and Yoon 2000 supported oral absorption, repeated-dose behavior, and CYP2D6 activity-score-dependent clearance. Verification used independent immediate-release and controlled-release oral studies, including Chen 2015, McClelland 1984, Schoedel 2012, Segura 2005, van der Lee 2007, and Yasui-Furukori 2006.

The model quantifies CYP2D6- and CYP3A4-mediated metabolism, unspecific hepatic clearance, and passive renal filtration. The CYP2D6 activity-score implementation is the major driver for differences between poor-, intermediate-, extensive-, and higher-activity groups. CYP3A4 metabolism and unspecific hepatic clearance provide additional clearance capacity.

The next sections show:

1. the final model input parameters for the building blocks: Section 3.1.
2. the overall goodness of fit: Section 3.2.
3. simulated vs. observed concentration-time profiles for the clinical studies used for model building and for model verification: Section 3.3.

The concentration-time profiles should be interpreted by study context. Intravenous profiles mainly test distribution and systemic clearance. Immediate-release oral profiles test the combined absorption and first-pass components. Controlled-release profiles additionally test the Weibull dissolution parameters. CYP2D6 poor-metabolizer and low-activity-score profiles are particularly informative for CYP2D6-independent clearance.

3.1 Paroxetine final input parameters

The following tables summarize the final input parameters for paroxetine. Interpret optimized parameters together with the sources in Section 2.2 and the assumptions in Section 2.3.

Compound: Paroxetine

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	7.31 mg/ml	Publication-In Vitro-Khatavkar et al. 2013	Khatavkar 2013 (pH 4.5)	True
Reference pH	4.5	Publication-In Vitro-Khatavkar et al. 2013	Khatavkar 2013 (pH 4.5)	True
Lipophilicity	3.95 Log Units	Publication-Other-Agrawal 2013	optimized	True
Lipophilicity	3.95 Log Units		Agrawal 2013	False
Lipophilicity	3.15 Log Units		ChemAxon	False
Lipophilicity	3.4 Log Units		Zhong 2014	False
Fraction unbound (plasma, reference value)	0.0487321988	Parameter Identification-Parameter Identification-Optimized	Optimized	True
Specific intestinal permeability (transcellular)	3.93E-05 cm/min	Parameter Identification-Parameter Identification-Optimized	Optimized	True
F	1			
Is small molecule	Yes			
Molecular weight	329.37 g/mol			
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Rodgers and Rowland
Cellular permeabilities	PK-Sim Standard

Processes

Metabolizing Enzyme: CYP2D6-Jornil 2010

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax/recombinant enzyme	9.7 pmol/min/pmol rec. enzyme	Publication-In Vitro-Jornil et al. 2010
Km	0.028 µmol/l	Publication-In Vitro-Jornil et al. 2010
kcat	1.3693020782 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP3A4-Jornil 2010

Molecule: CYP3A4

Parameters

Name	Value	Value Origin
In vitro Vmax/recombinant enzyme	5.32 pmol/min/pmol rec. enzyme	Publication-In Vitro-Jornil et al. 2010
Km	4.7 µmol/l	Publication-In Vitro-Jornil et al. 2010
kcat	1.01 1/min	Parameter Identification-Parameter Identification-Optimized

Systemic Process: Glomerular Filtration-Assumption

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	1	Other-Assumption-Assumed

Inhibition: CYP2D6-Obach, 2007

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
kinact	0.17 1/min	Publication-In Vitro-Obach et al. 2007
K_kinact_half	0.16738 µmol/l	Parameter Identification-Parameter Identification-Optimized

Inhibition: CYP3A4-Obach 2007

Molecule: CYP3A4

Parameters

Name	Value	Value Origin
kinact	0.011 1/min	Publication-In Vitro-Obach et al. 2007
K_kinact_half	4.48 µmol/l	Publication-In Vitro-Obach et al. 2007

Metabolizing Enzyme: CYP2D6-AS=2

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax/recombinant enzyme	9.7 pmol/min/pmol rec. enzyme	
Km	0.028 µmol/l	Publication-In Vitro-Jornil et al. 2010
kcat	4.09 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=1.5

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax/recombinant enzyme	9.7 pmol/min/pmol rec. enzyme	
Km	0.028 µmol/l	Publication-In Vitro-Jornil et al. 2010
kcat	2.71 1/min	Other-Manual Fit-AS Model

Metabolizing Enzyme: CYP2D6-AS=1.25

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax/recombinant enzyme	9.7 pmol/min/pmol rec. enzyme	
Km	0.028 µmol/l	Publication-In Vitro-Jornil et al. 2010
kcat	2.11 1/min	Other-Manual Fit-AS-Model

Metabolizing Enzyme: CYP2D6-AS=1

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax/recombinant enzyme	9.7 pmol/min/pmol rec. enzyme	
Km	0.028 µmol/l	Publication-In Vitro-Jornil et al. 2010
kcat	1.56 1/min	Other-Manual Fit-AS Model

Metabolizing Enzyme: CYP2D6-AS=0.5

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax/recombinant enzyme	9.7 pmol/min/pmol rec. enzyme	
Km	0.028 µmol/l	Publication-In Vitro-Jornil et al. 2010
kcat	0.66 1/min	Other-Manual Fit-AS Model

Metabolizing Enzyme: CYP2D6-AS=0

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax/recombinant enzyme	9.7 pmol/min/pmol rec. enzyme	
Km	0.028 µmol/l	
kcat	0 1/min	Other-Assumption-Assumed

Metabolizing Enzyme: CYP2D6-AS=3

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax/recombinant enzyme	9.7 pmol/min/pmol rec. enzyme	
Km	0.028 µmol/l	Publication-In Vitro-Jornil et al. 2010
kcat	7.58 1/min	Other-Manual Fit-AS Model

Metabolizing Enzyme: CYP2D6-AS=0.75

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax/recombinant enzyme	9.7 pmol/min/pmol rec. enzyme	
Km	0.028 µmol/l	Publication-In Vitro-Jornil et al. 2010
kcat	1.08 1/min	Other-Manual Fit-AS Model

Systemic Process: Total Hepatic Clearance-Assumption

Species: Human

Parameters

Name	Value	Value Origin
Fraction unbound (experiment)	0.0487321988	
Lipophilicity (experiment)	3.95 Log Units	
Plasma clearance	0 ml/min/kg	
Specific clearance	2.2538632474 1/min	Parameter Identification-Parameter Identification-Optimized

3.2 Diagnostic plots

The goodness-of-fit diagnostics combine all paroxetine observations in one set of plots. The plot does not split model-building and model-verification observations. Use the concentration-time profile sections for that comparison.

Table 3-1: GMFE for Paroxetine goodness-of-fit diagnostics

Group	GMFE
Paroxetine	1.47

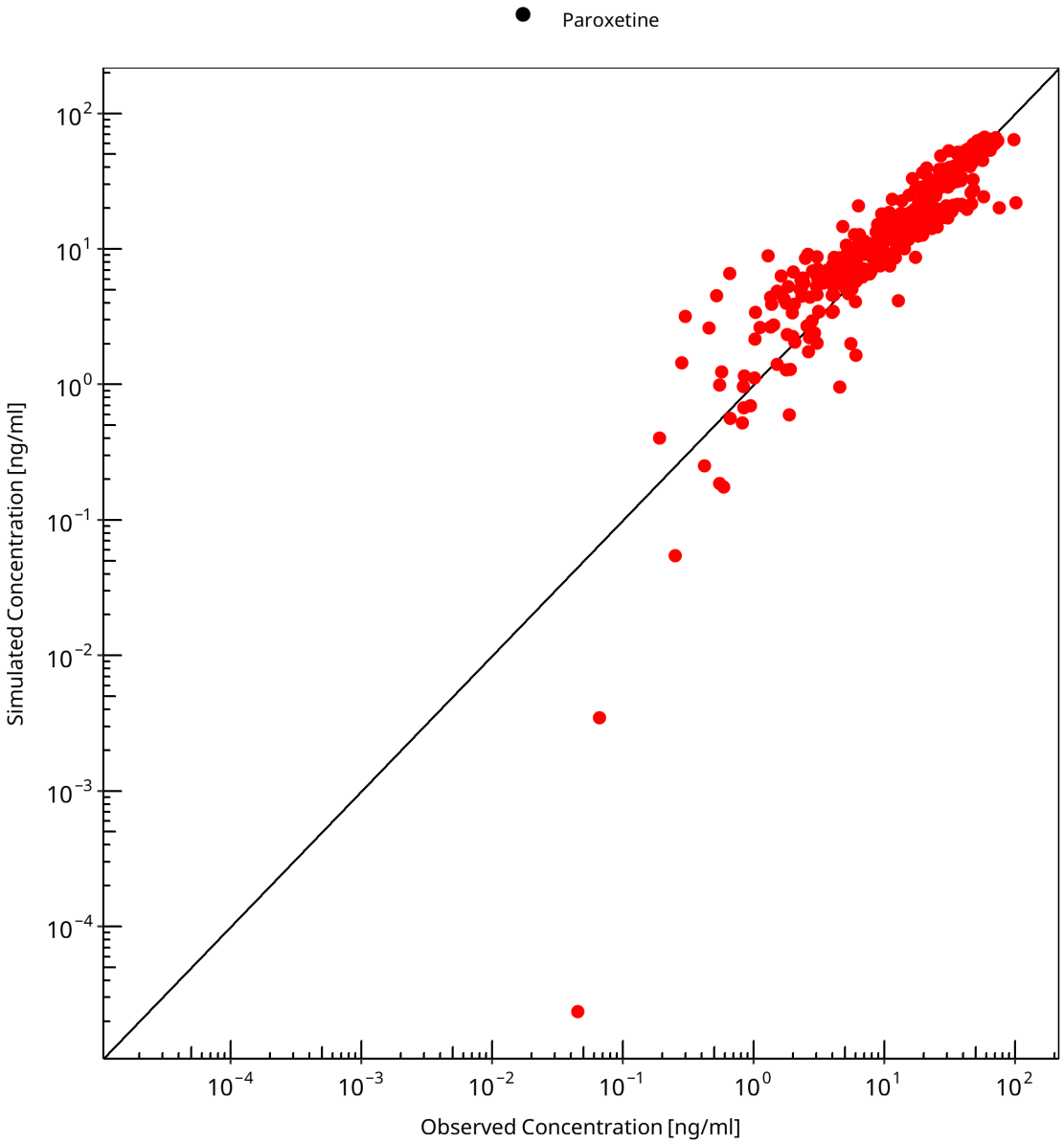


Figure 3-1: Paroxetine goodness-of-fit diagnostics

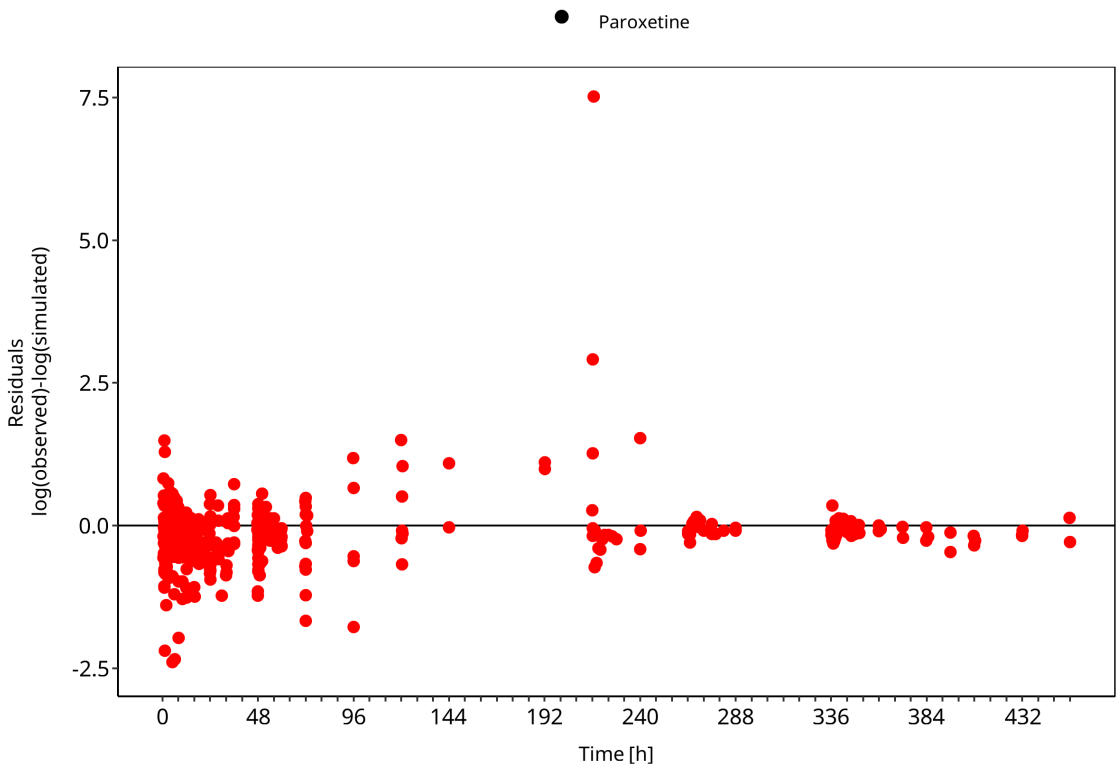


Figure 3-2: Paroxetine goodness-of-fit diagnostics

3.3 Concentration-time profiles

Simulated versus observed concentration-time profiles of all data listed in Section 2.2.2 are presented below.

3.3.1 Model building

The concentration-time profiles in this section show the clinical study arms used for model building. These profiles support assessment of systemic disposition, oral absorption, repeated-dose behavior, and CYP2D6 activity-score-dependent paroxetine exposure.

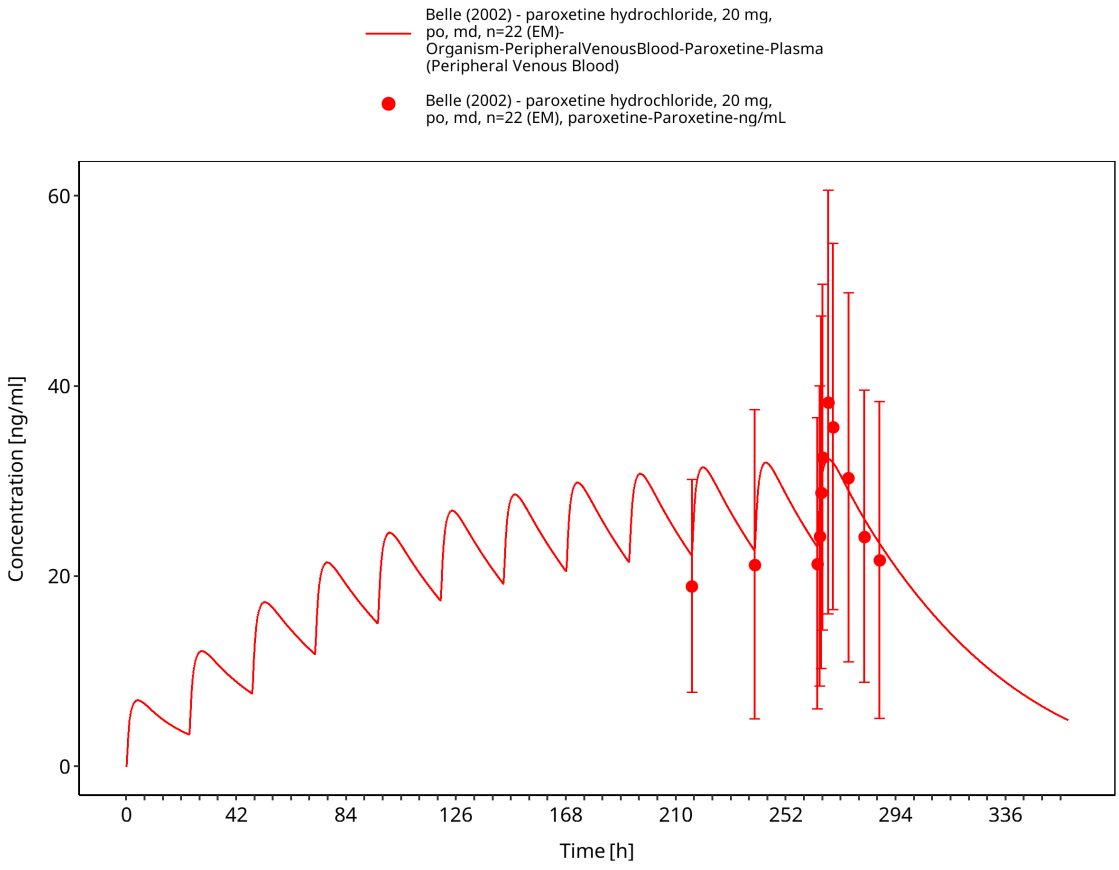


Figure 3-3: Belle 2002: paroxetine hydrochloride, 20 mg, po, md, n=22 (EM)

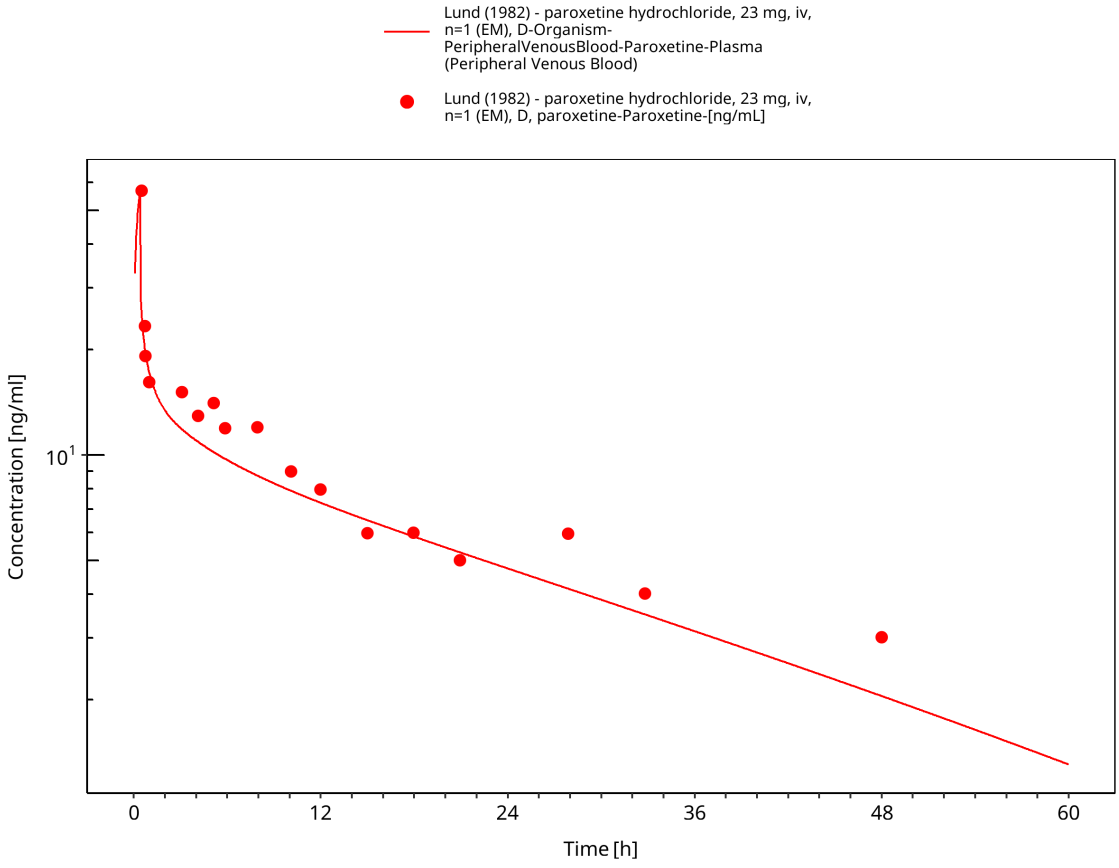


Figure 3-4: Lund 1982: paroxetine hydrochloride, 23 mg, iv, n=1 (EM), D

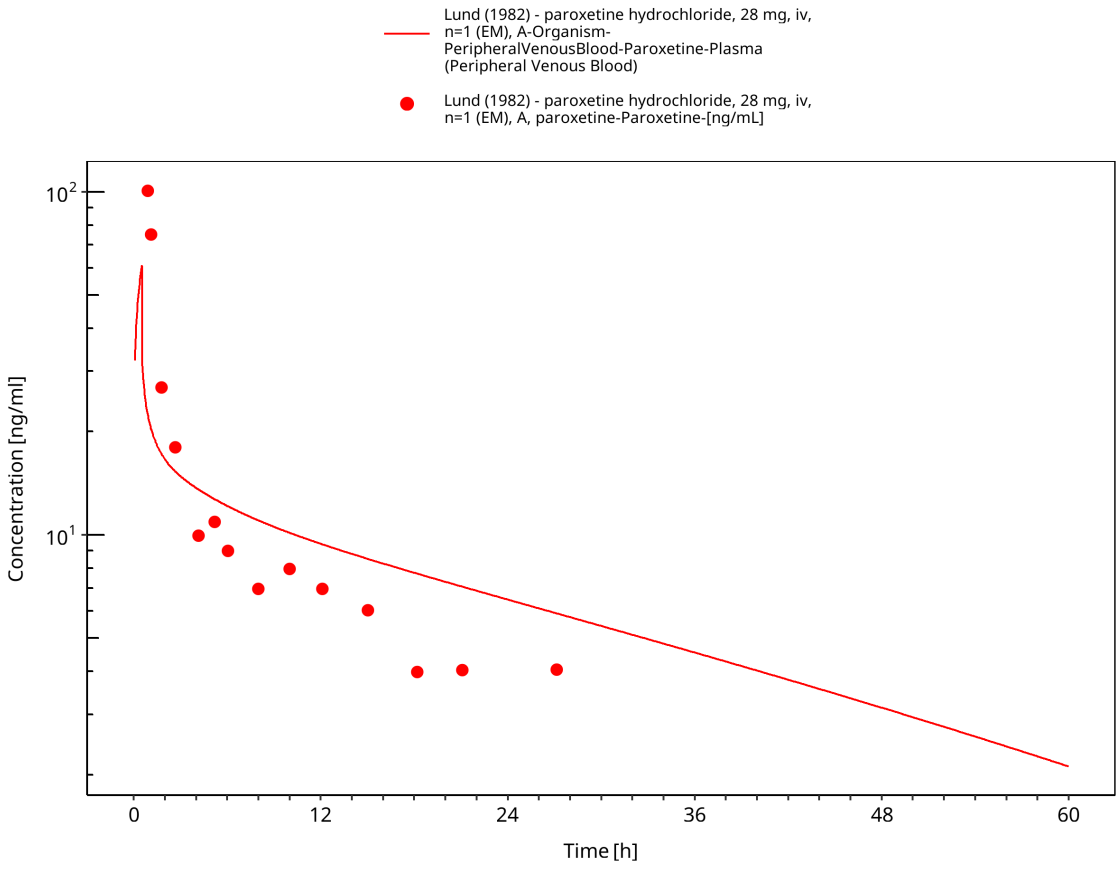


Figure 3-5: Lund 1982: paroxetine hydrochloride, 28 mg, iv, n=1 (EM), A

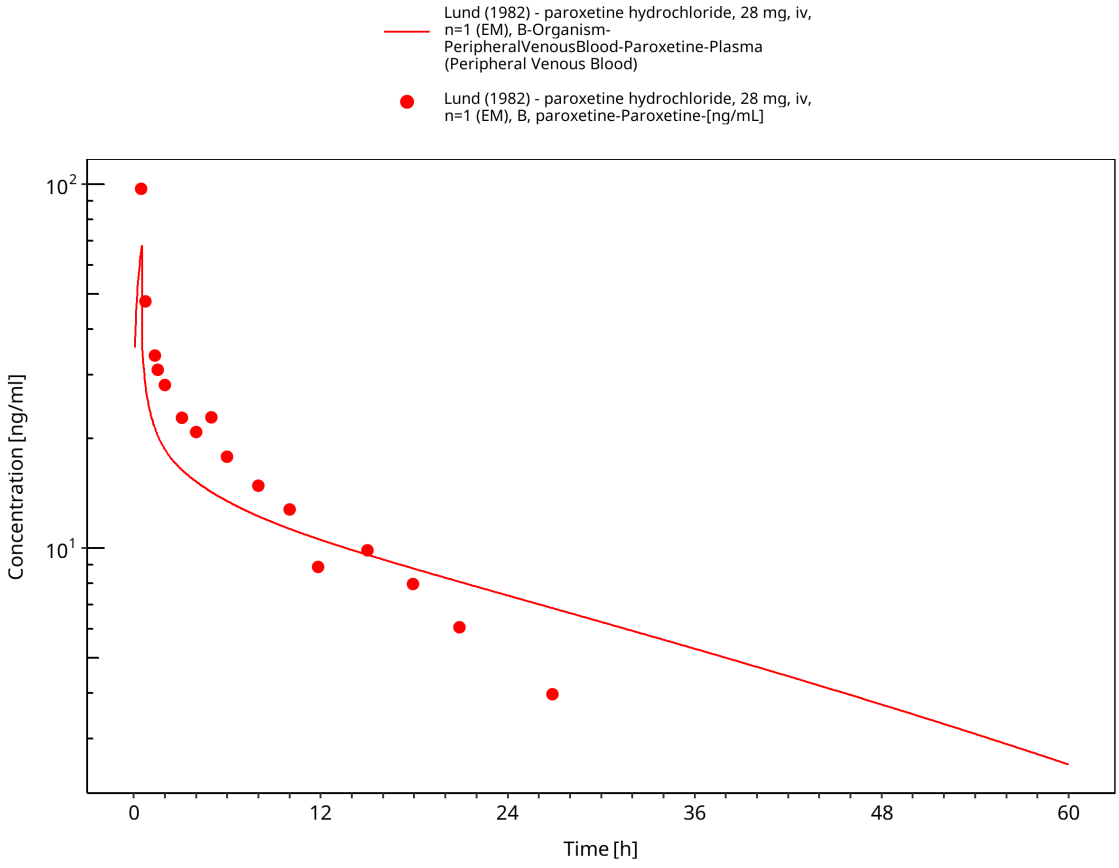


Figure 3-6: Lund 1982: paroxetine hydrochloride, 28 mg, iv, n=1 (EM), B

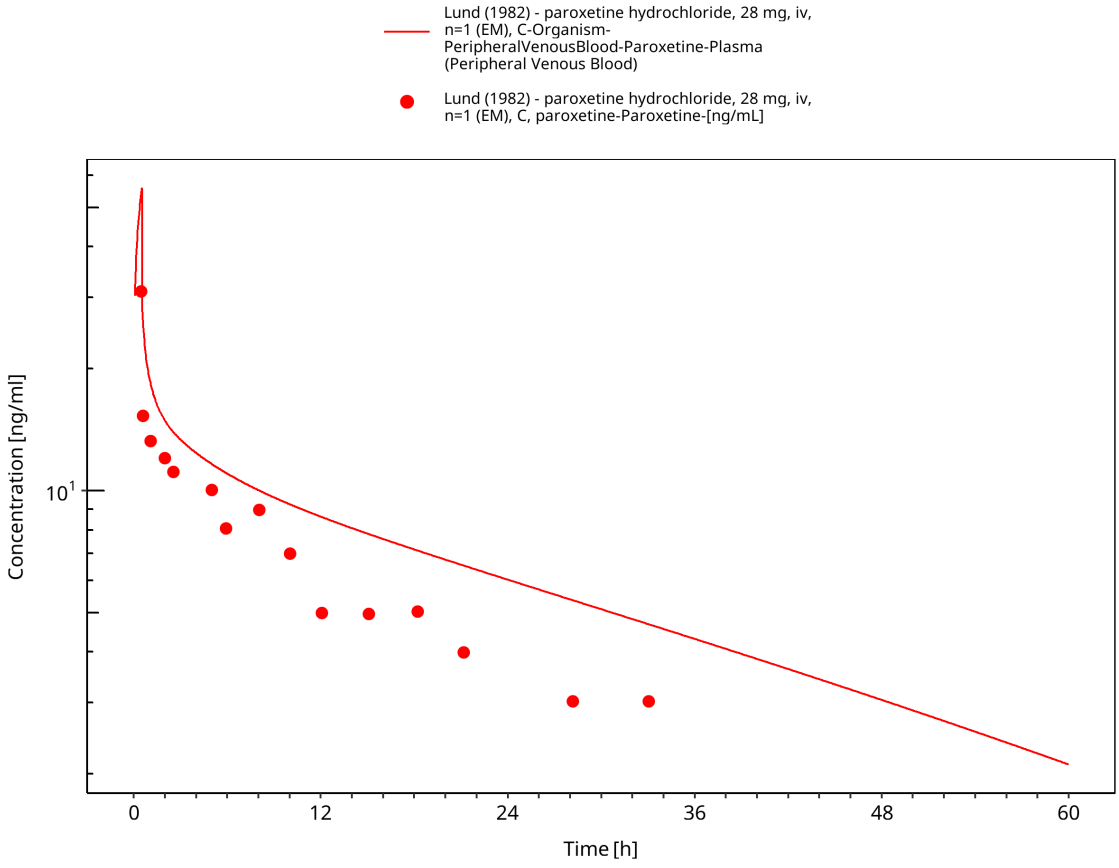


Figure 3-7: Lund 1982: paroxetine hydrochloride, 28 mg, iv, n=1 (EM), C

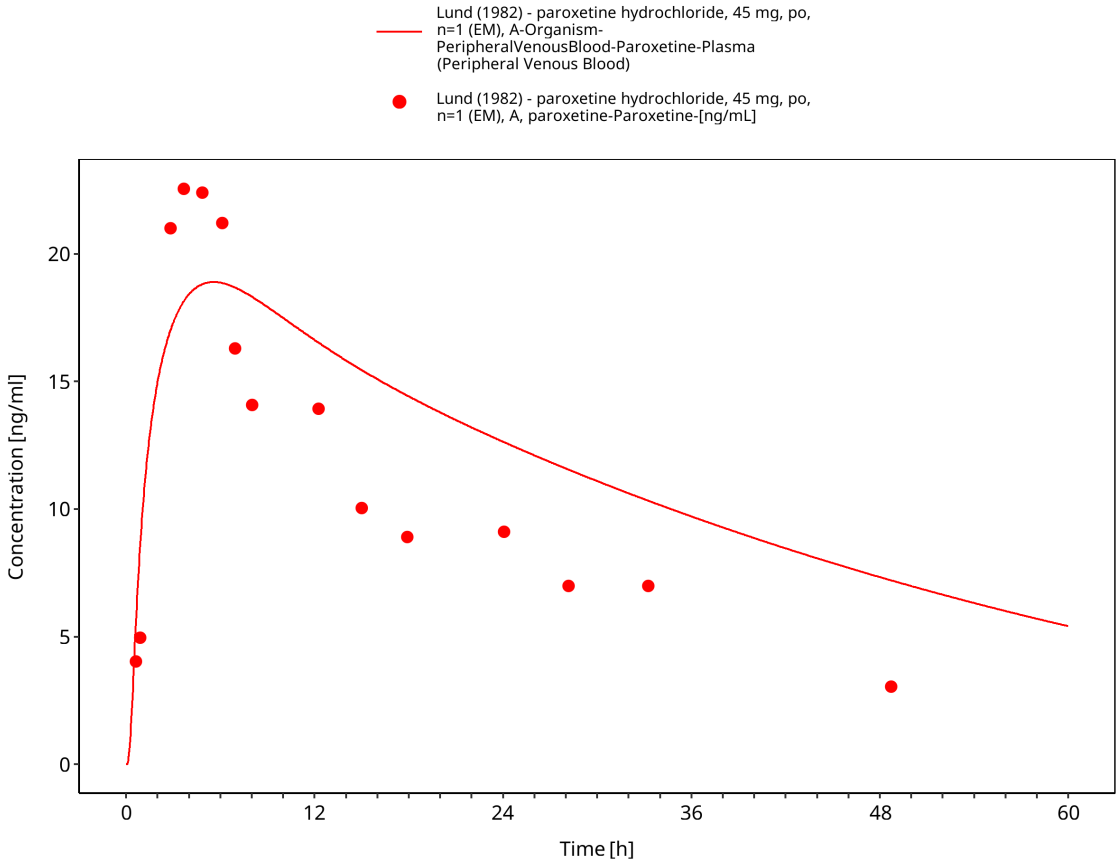


Figure 3-8: Lund 1982: paroxetine hydrochloride, 45 mg, po, n=1 (EM), A

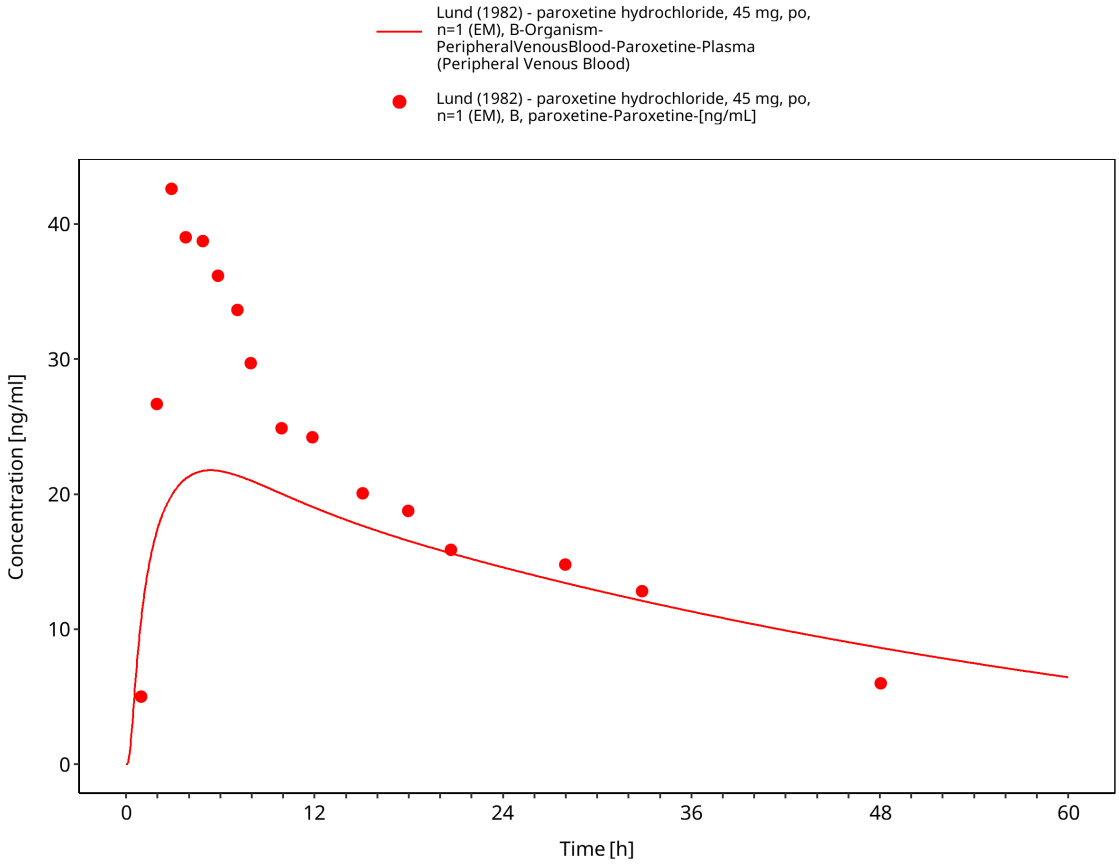


Figure 3-9: Lund 1982: paroxetine hydrochloride, 45 mg, po, n=1 (EM), B

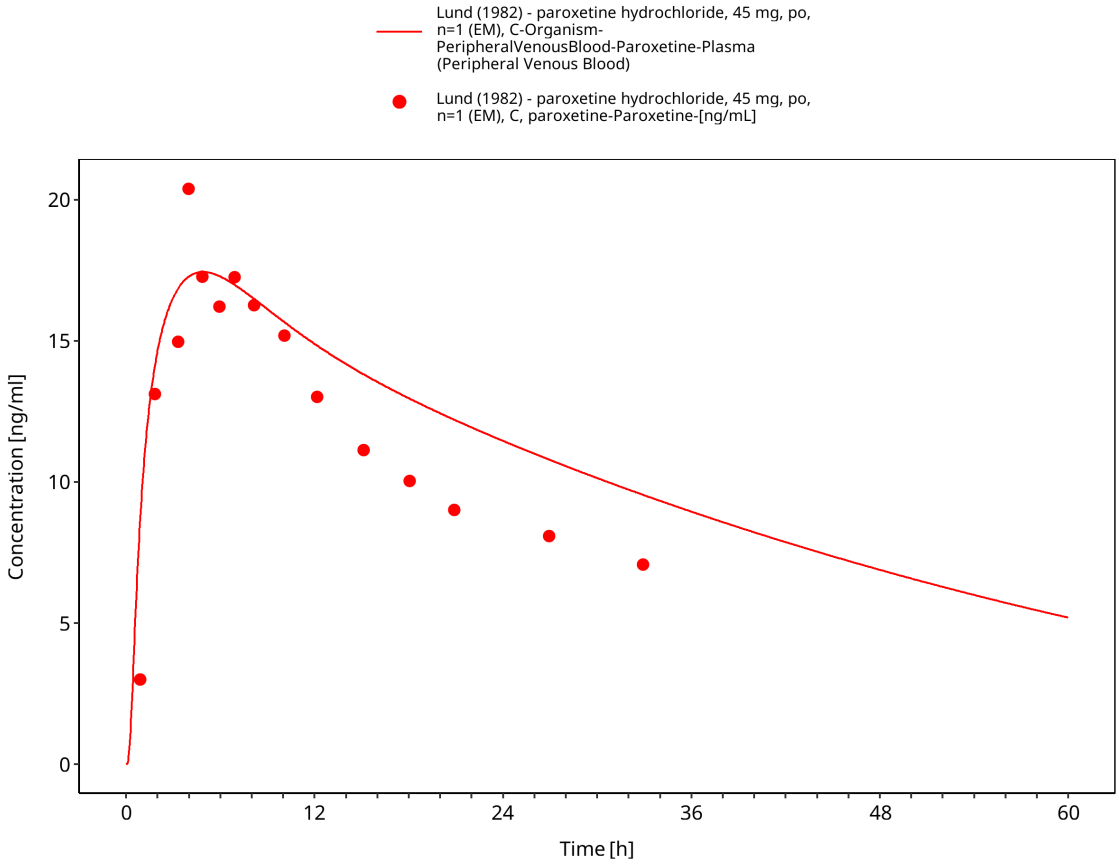


Figure 3-10: Lund 1982: paroxetine hydrochloride, 45 mg, po, n=1 (EM), C

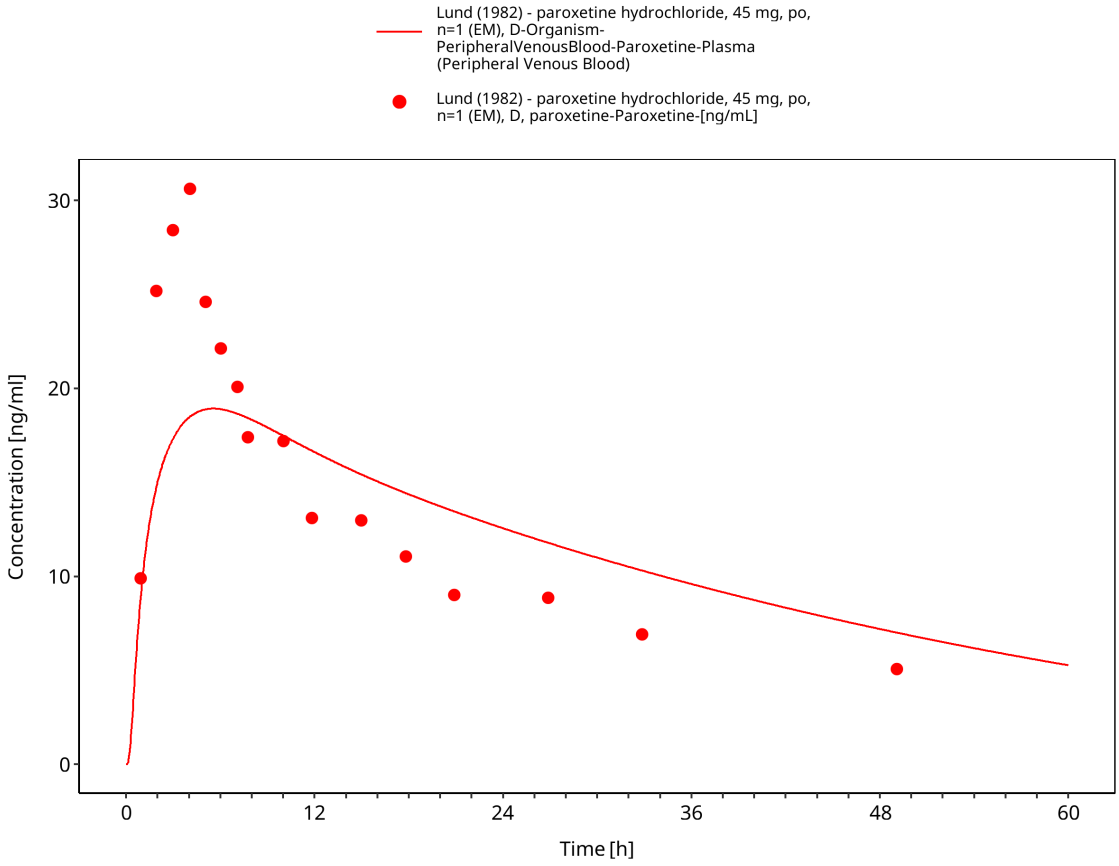


Figure 3-11: Lund 1982: paroxetine hydrochloride, 45 mg, po, n=1 (EM), D

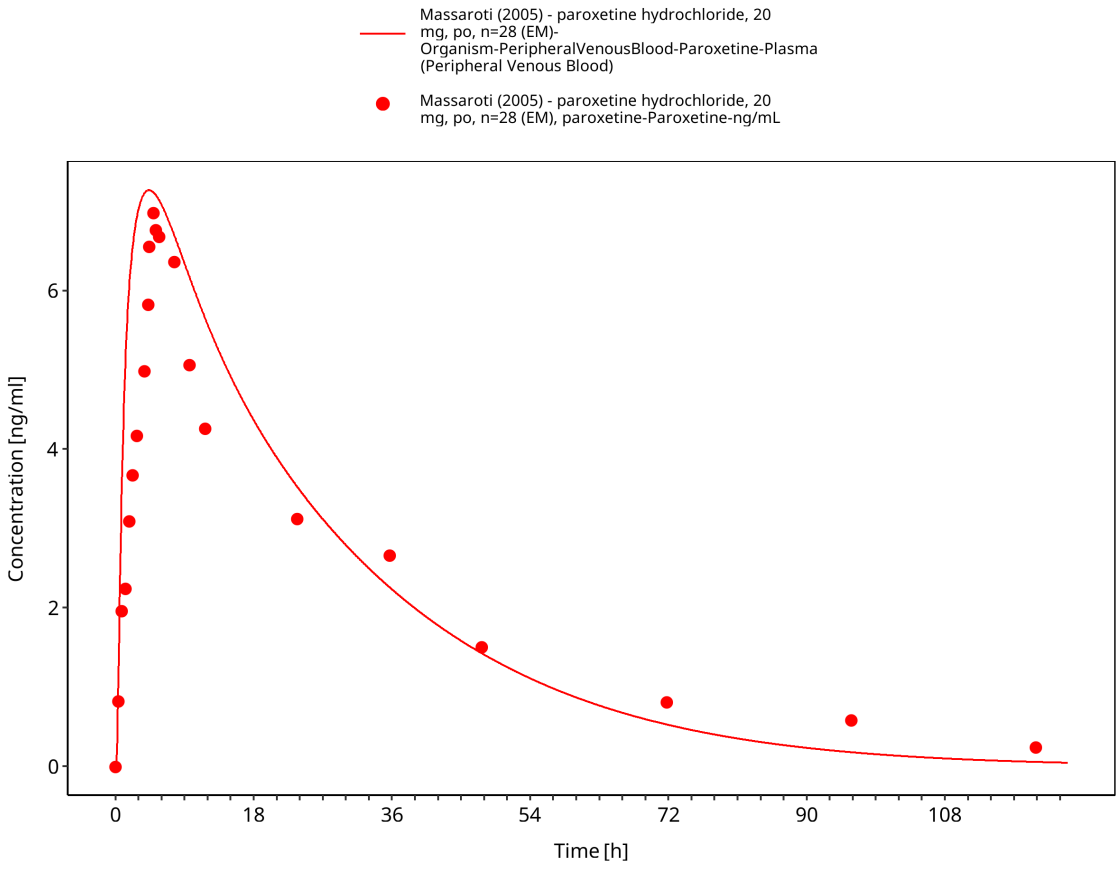


Figure 3-12: Massaroti 2005: paroxetine hydrochloride, 20 mg, po, n=28 (EM)

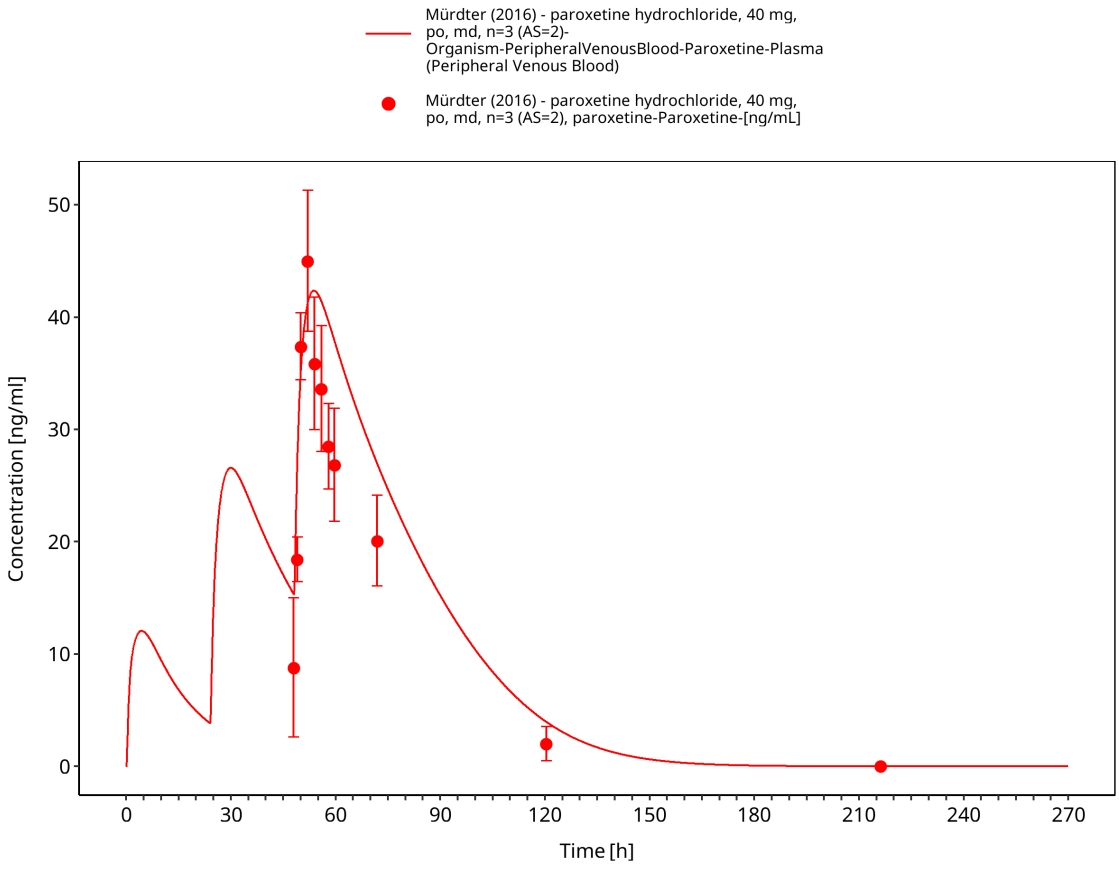


Figure 3-13: Mürdter 2016: paroxetine hydrochloride, 40 mg, po, md, n=3 (AS=2)

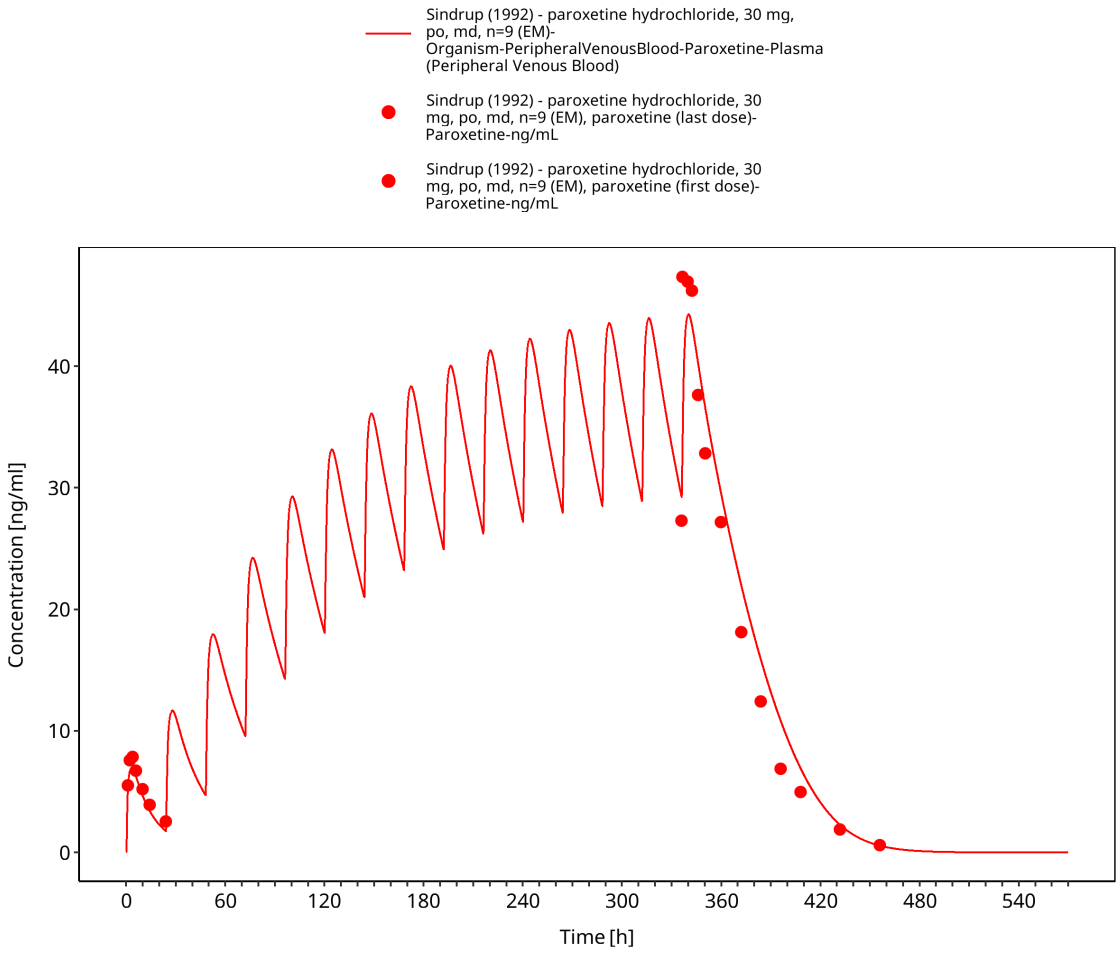


Figure 3-14: Sindrup 1992: paroxetine hydrochloride, 30 mg, po, md, n=9 (EM)

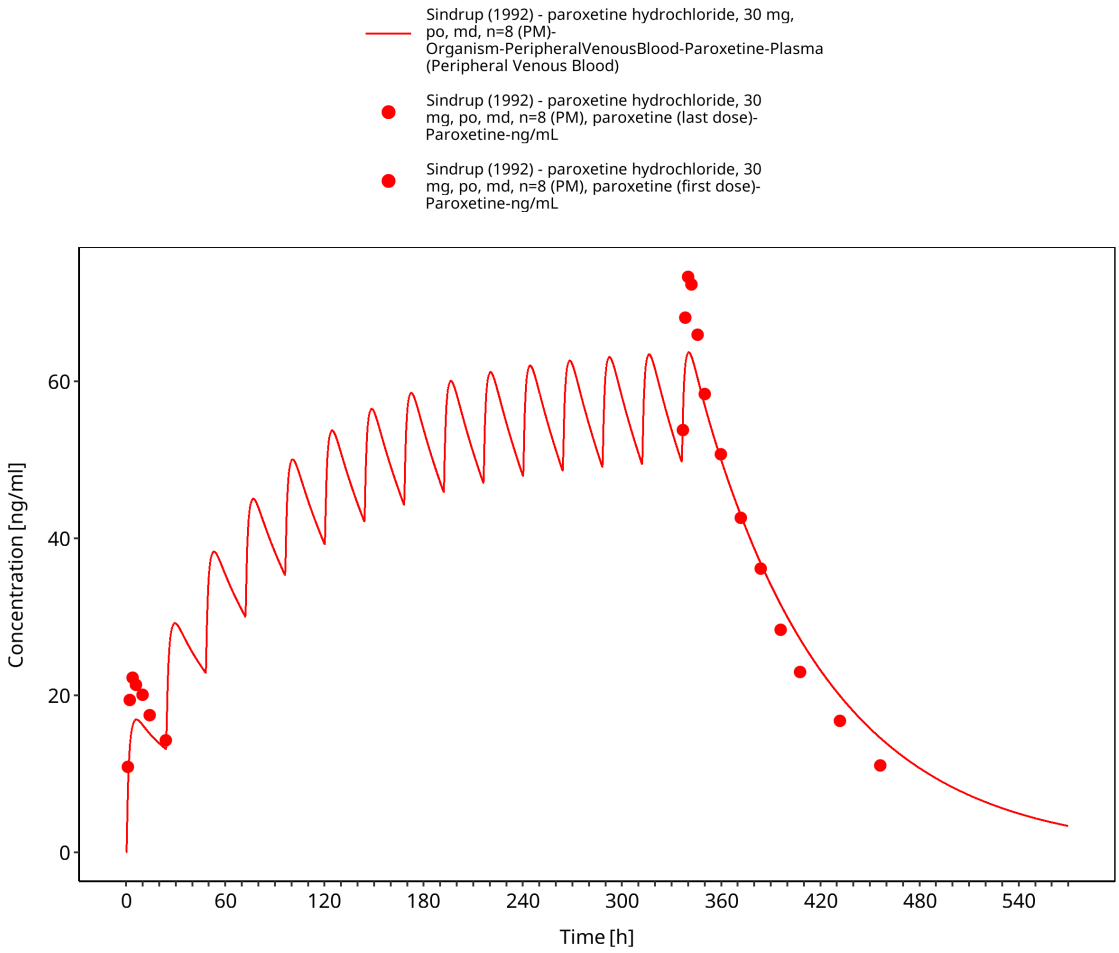


Figure 3-15: Sindrup 1992: paroxetine hydrochloride, 30 mg, po, md, n=8 (PM)

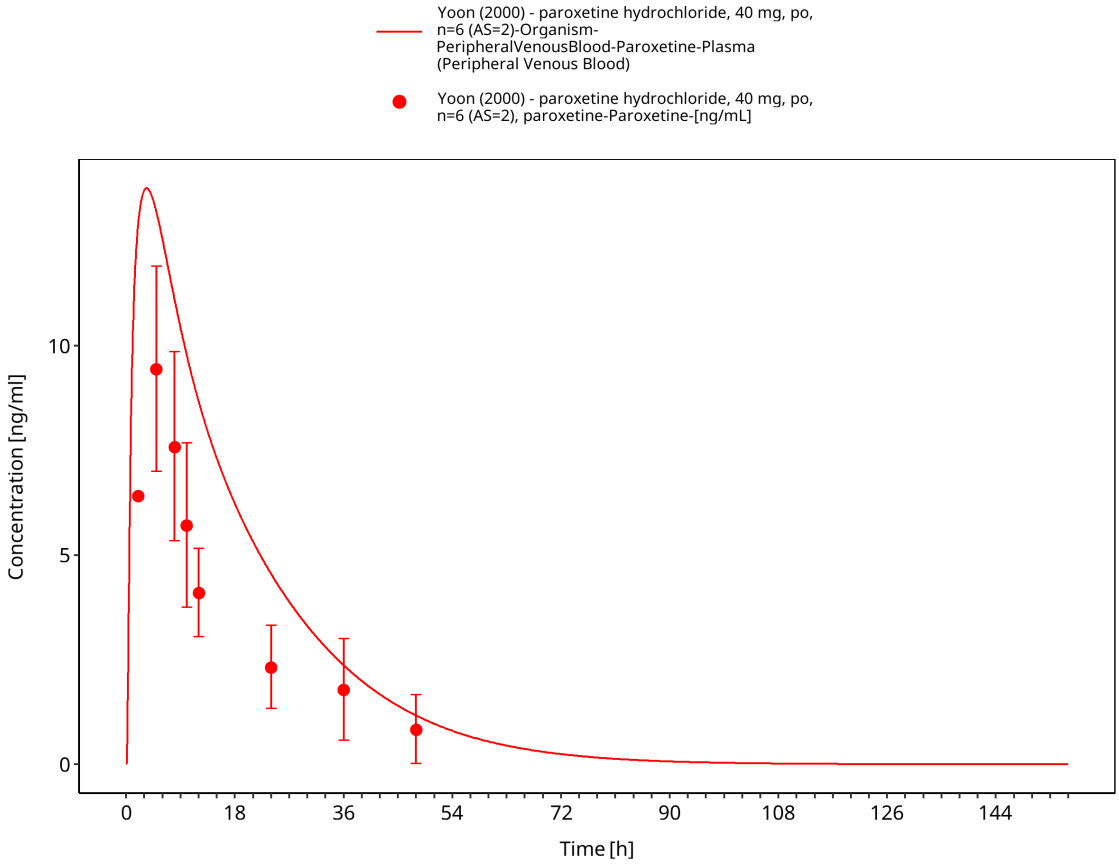


Figure 3-16: Yoon 2000: paroxetine hydrochloride, 40 mg, po, n=6 (AS=2)

3.3.2 Model verification

The concentration-time profiles in this section show independent clinical study arms used for model verification. These profiles support assessment of model performance across additional oral dosing regimens, formulations, and CYP2D6 activity-score groups.

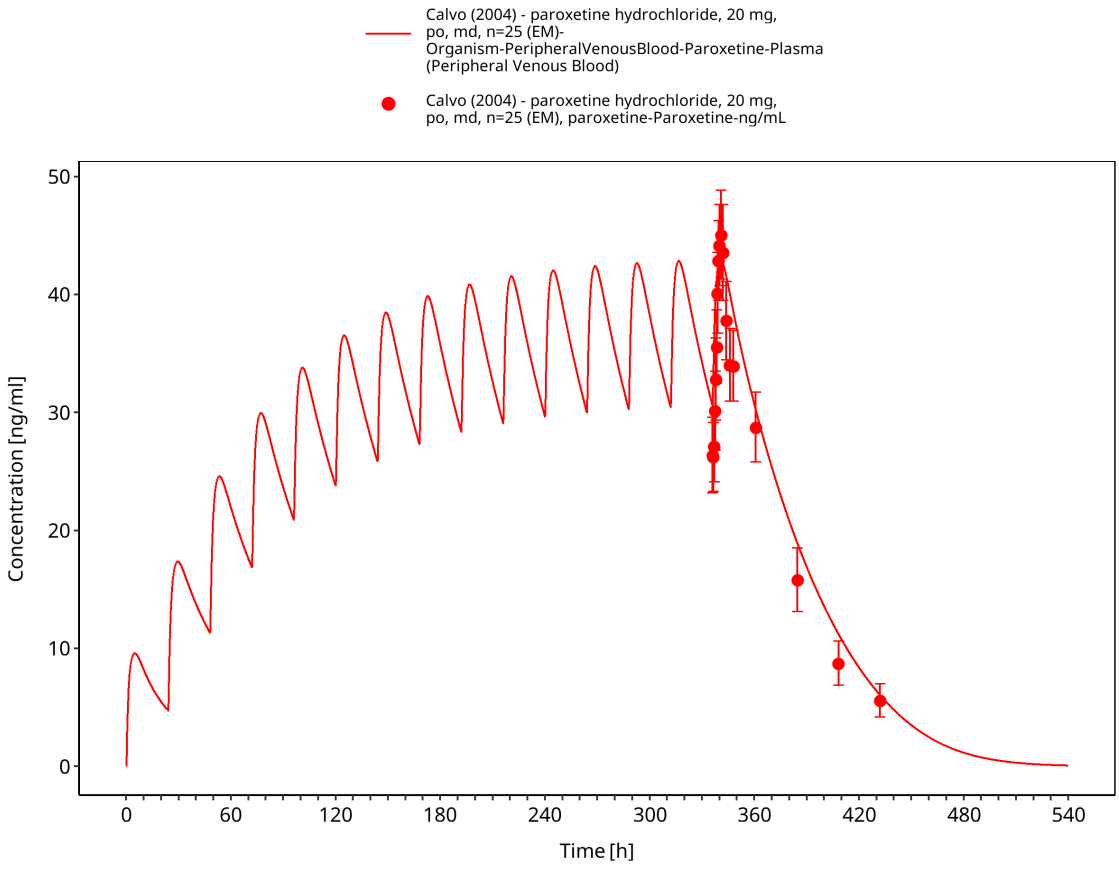


Figure 3-17: Calvo 2004: paroxetine hydrochloride, 20 mg, po, md, n=25 (EM)

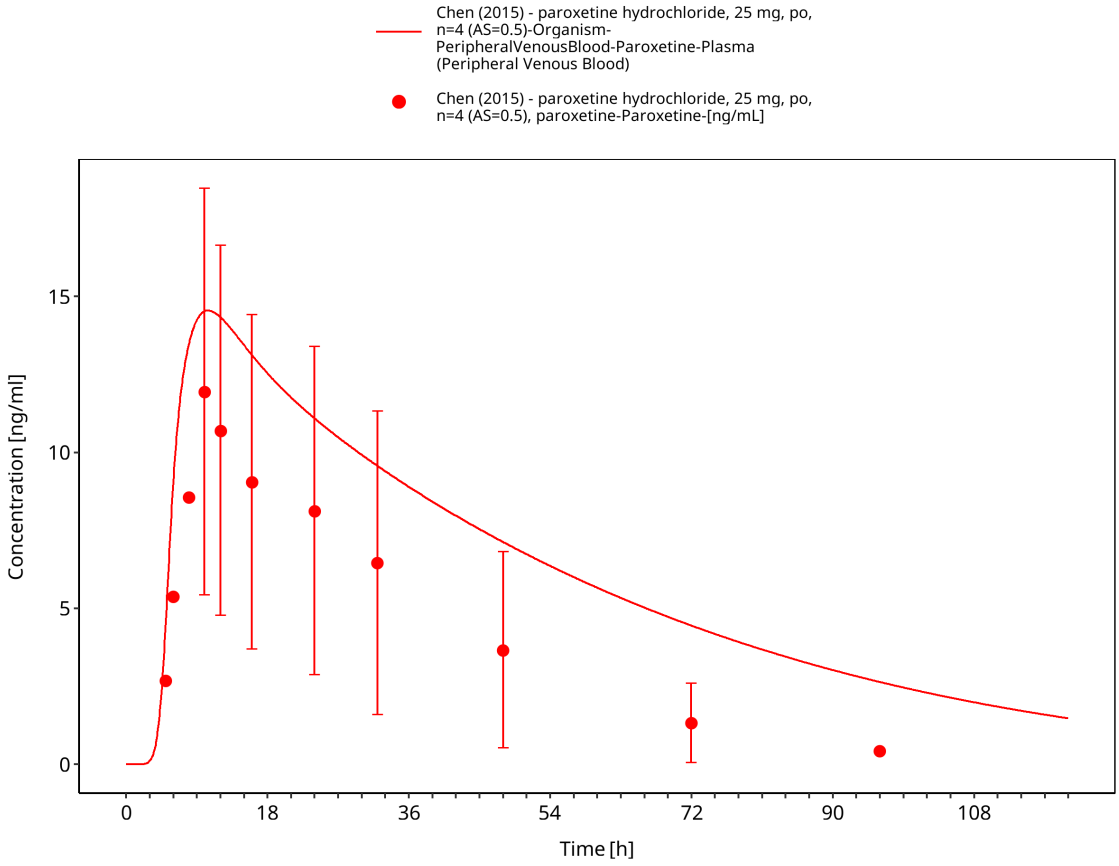


Figure 3-18: Chen 2015: paroxetine hydrochloride, 25 mg, po, n=4 (AS=0.5)

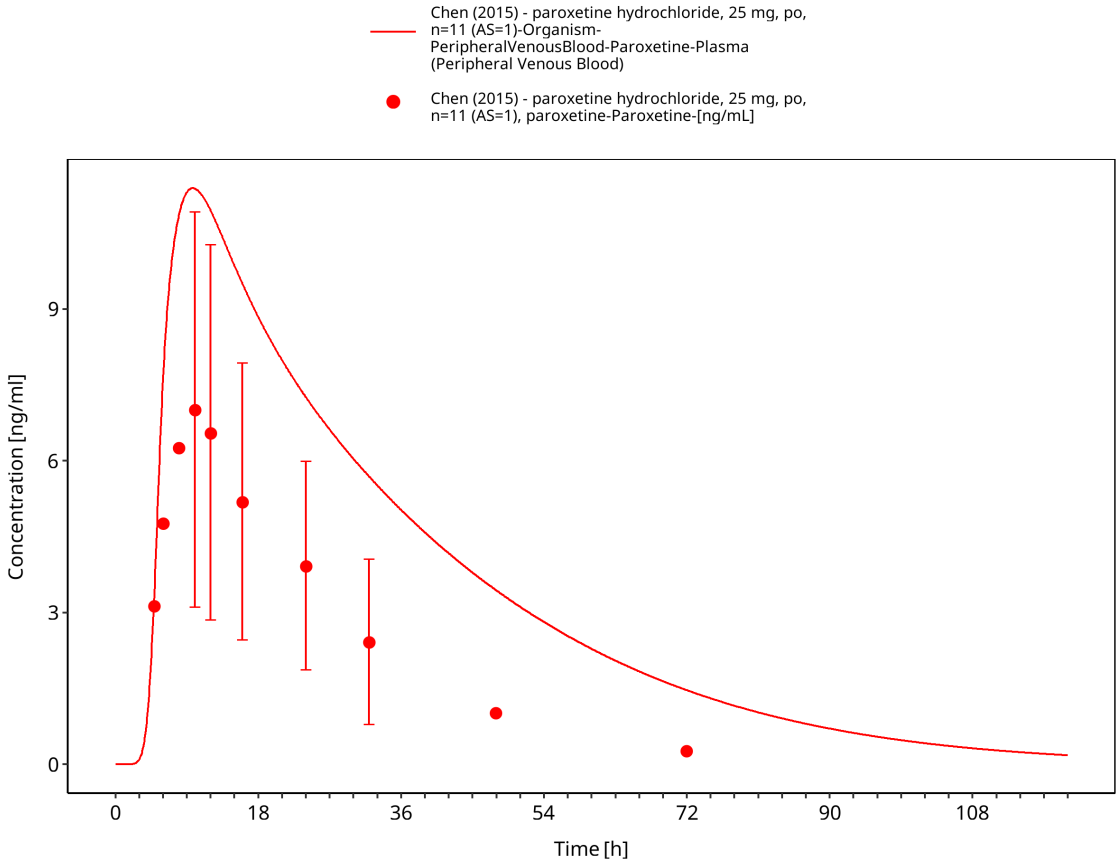


Figure 3-19: Chen 2015: paroxetine hydrochloride, 25 mg, po, n=11 (AS=1)

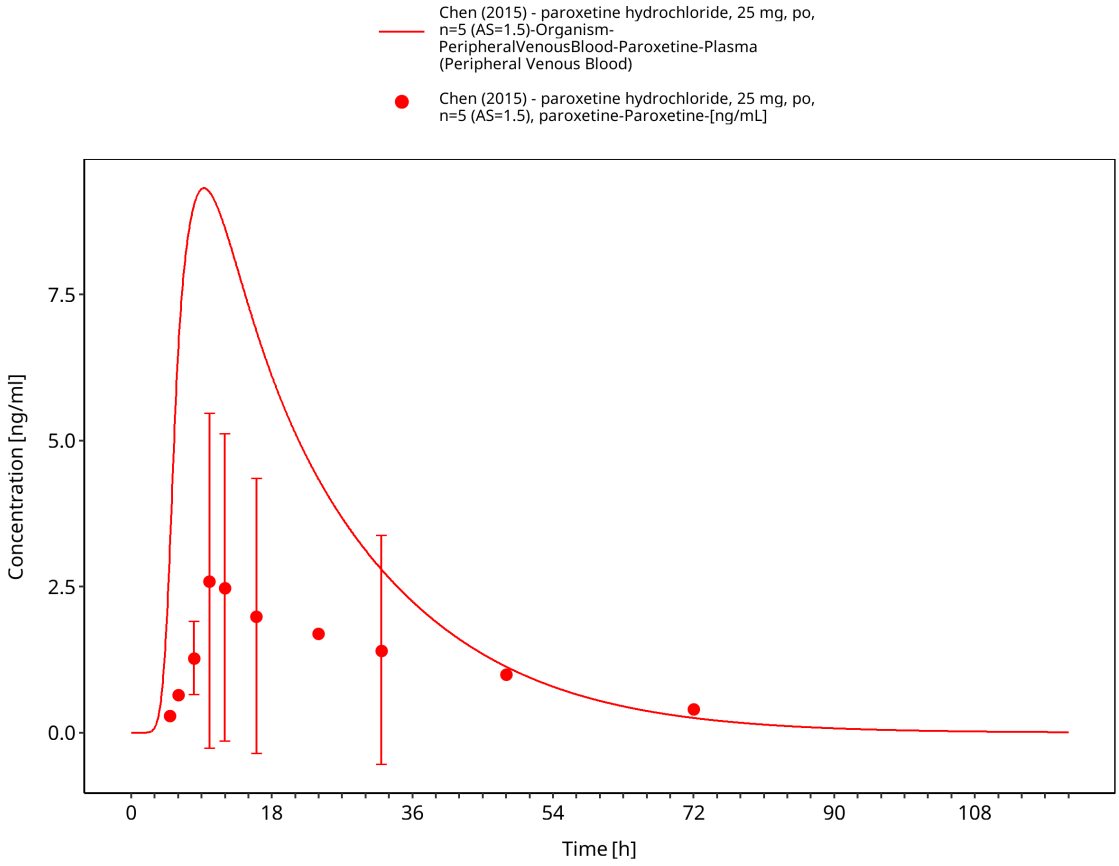


Figure 3-20: Chen 2015: paroxetine hydrochloride, 25 mg, po, n=5 (AS=1.5)

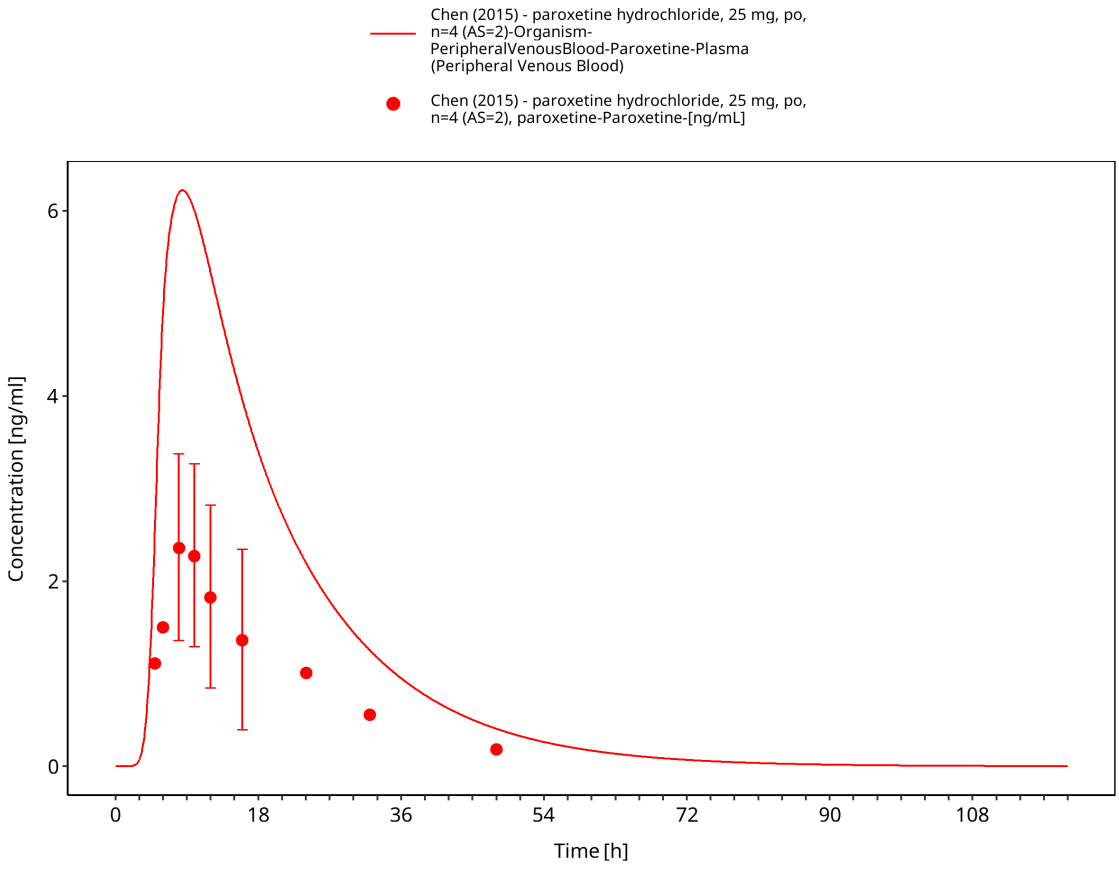


Figure 3-21: Chen 2015: paroxetine hydrochloride, 25 mg, po, n=4 (AS=2)

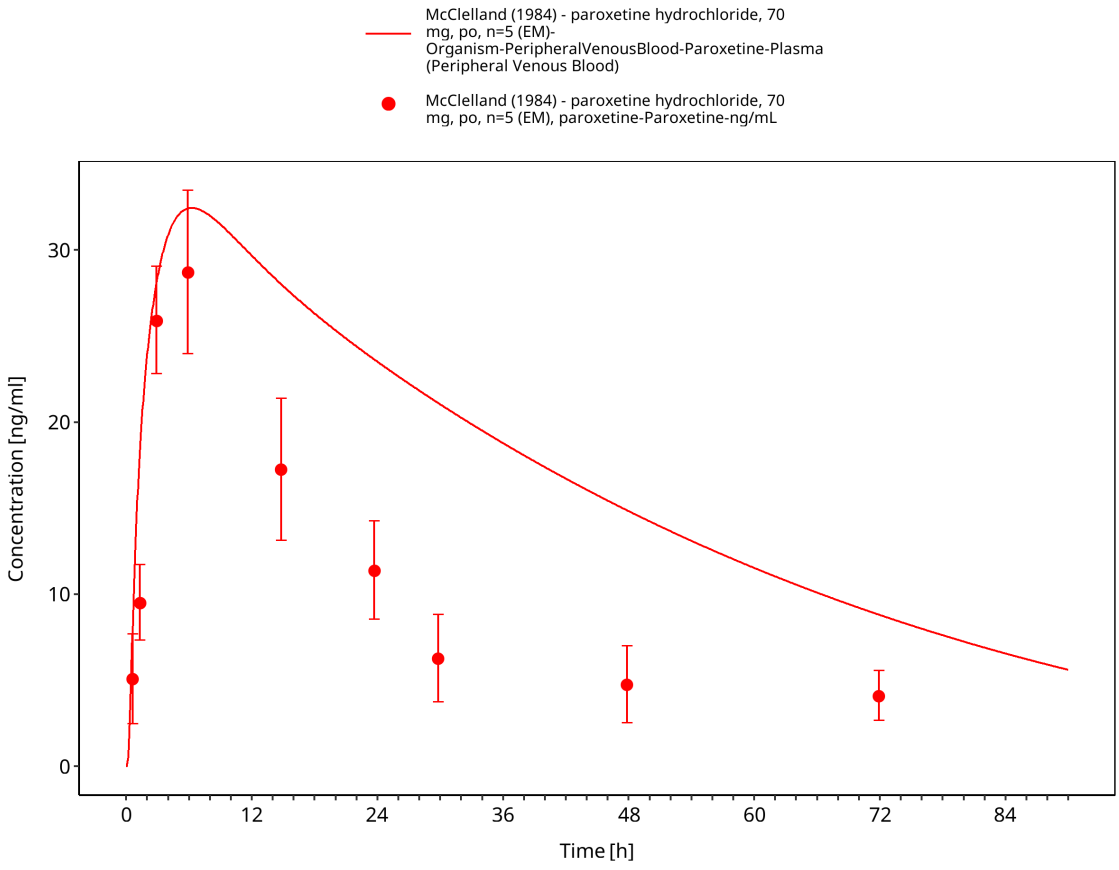


Figure 3-22: McClelland 1984: paroxetine hydrochloride, 70 mg, po, n=5 (EM)

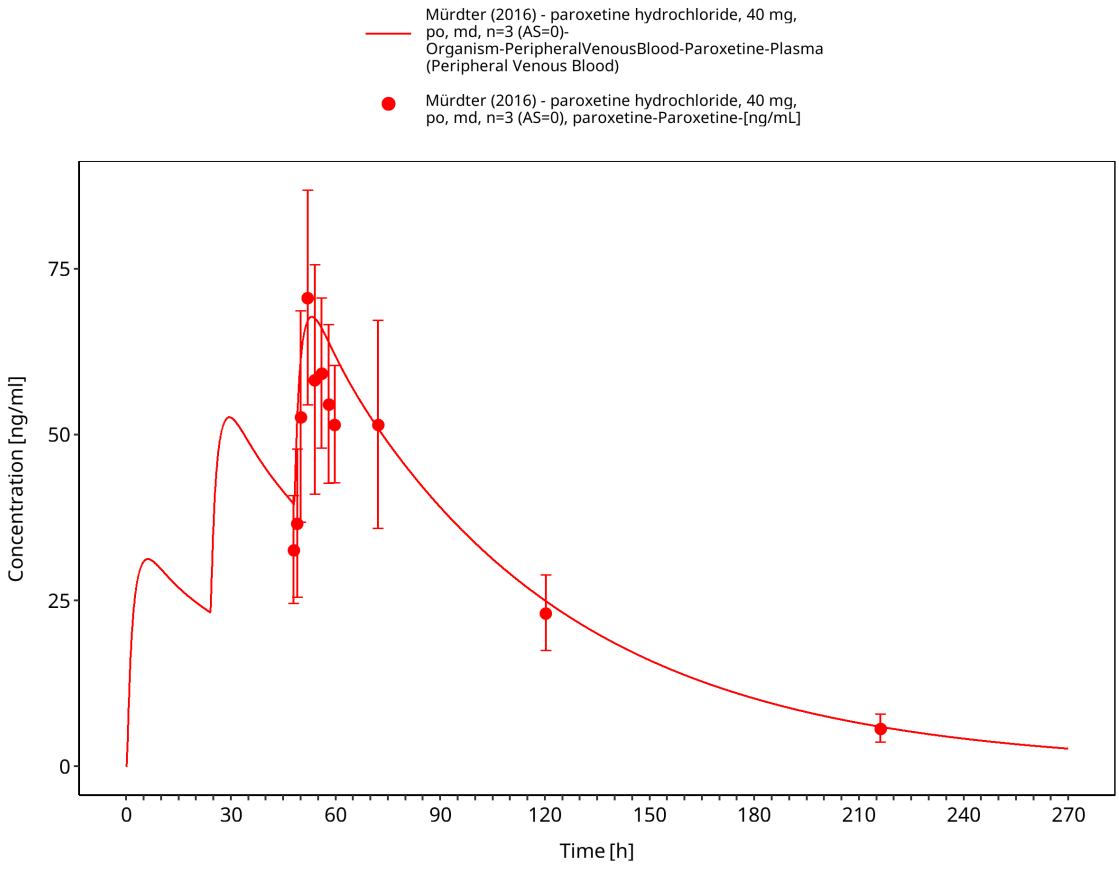


Figure 3-23: Mürdter 2016: paroxetine hydrochloride, 40 mg, po, md, n=3 (AS=0)

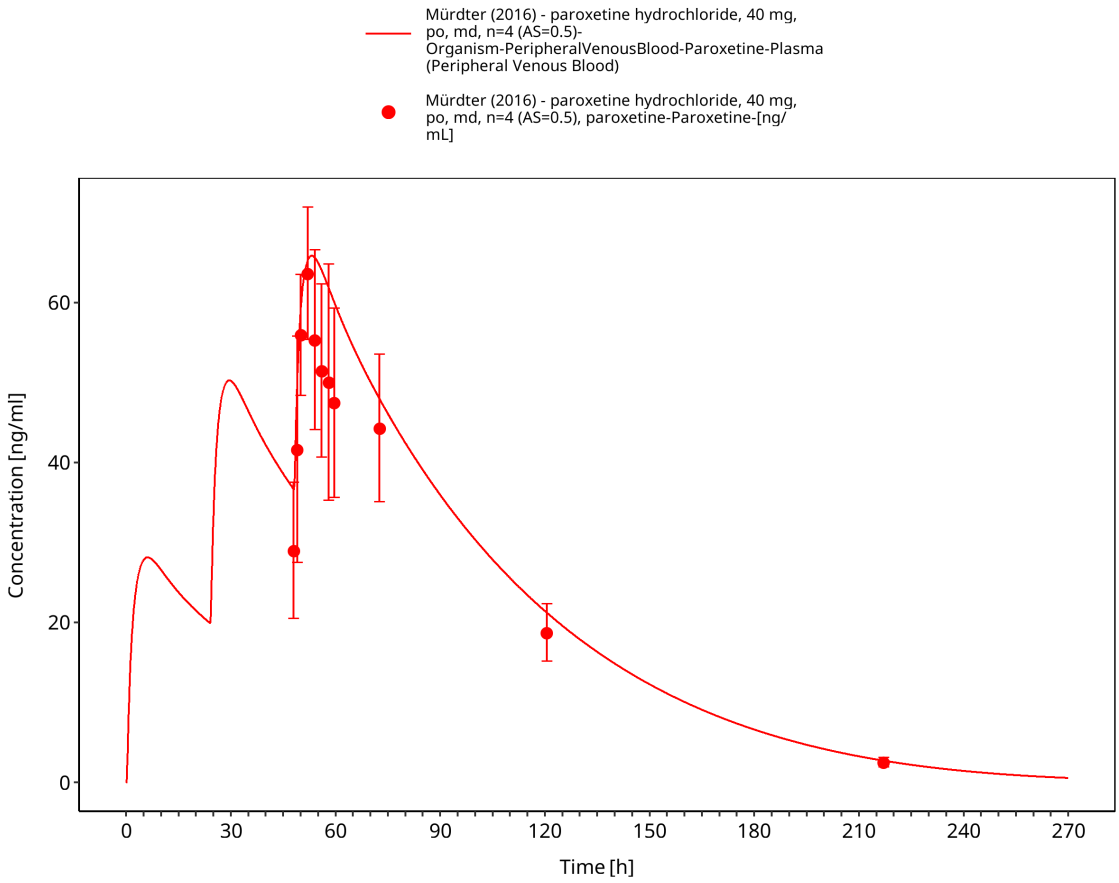


Figure 3-24: Mürdter 2016: paroxetine hydrochloride, 40 mg, po, md, n=4 (AS=0.5)

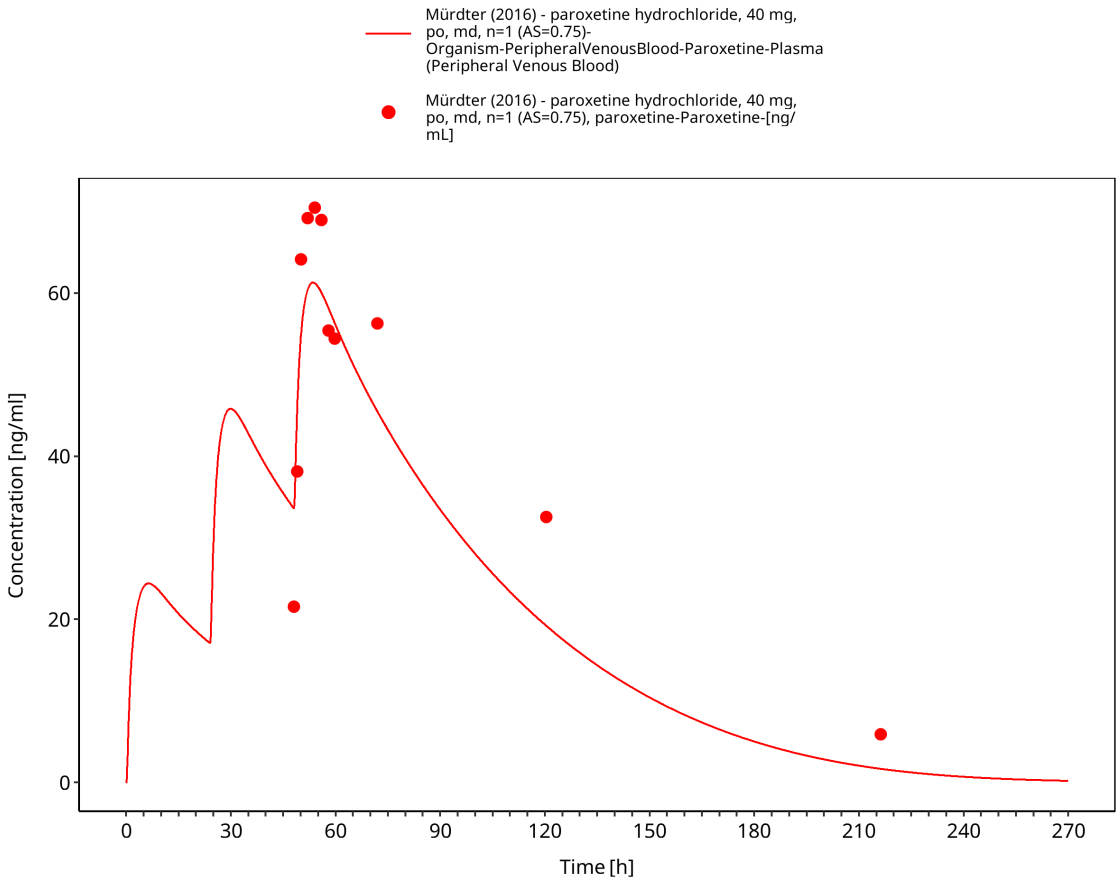


Figure 3-25: Mürdter 2016: paroxetine hydrochloride, 40 mg, po, md, n=1 (AS=0.75)

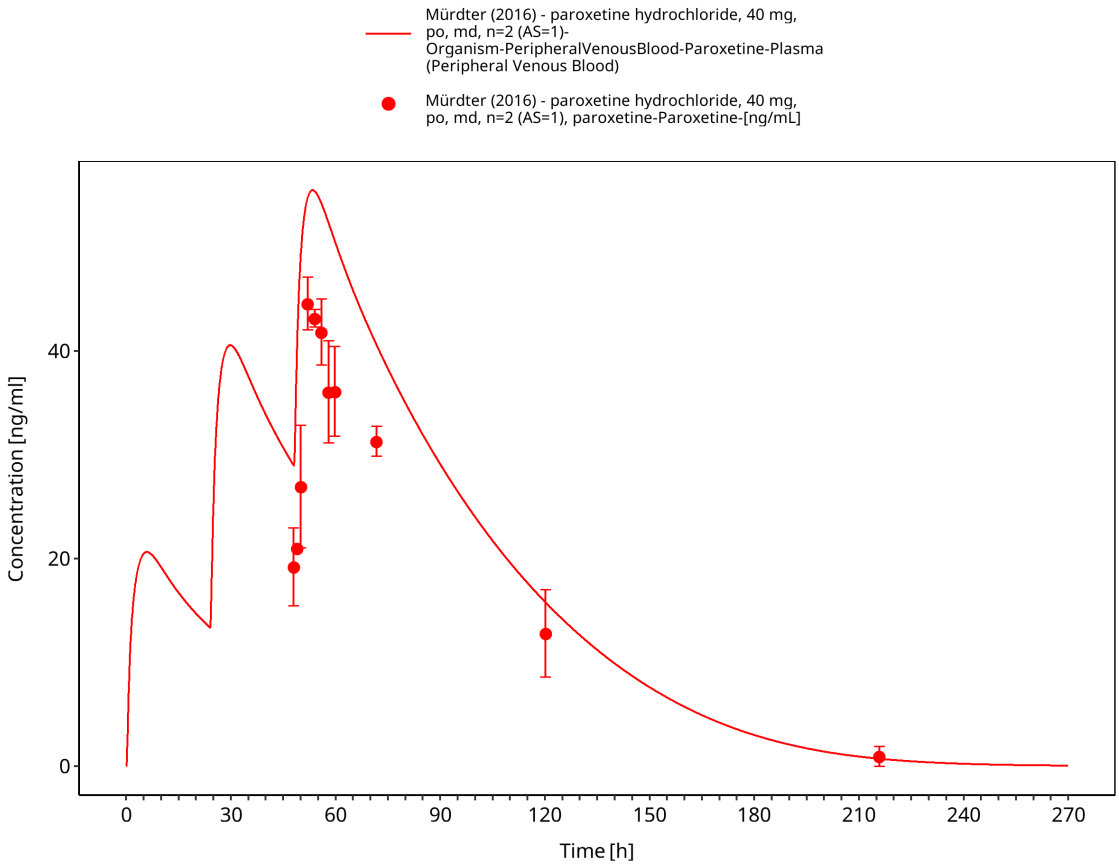


Figure 3-26: Mürdter 2016: paroxetine hydrochloride, 40 mg, po, md, n=2 (AS=1)

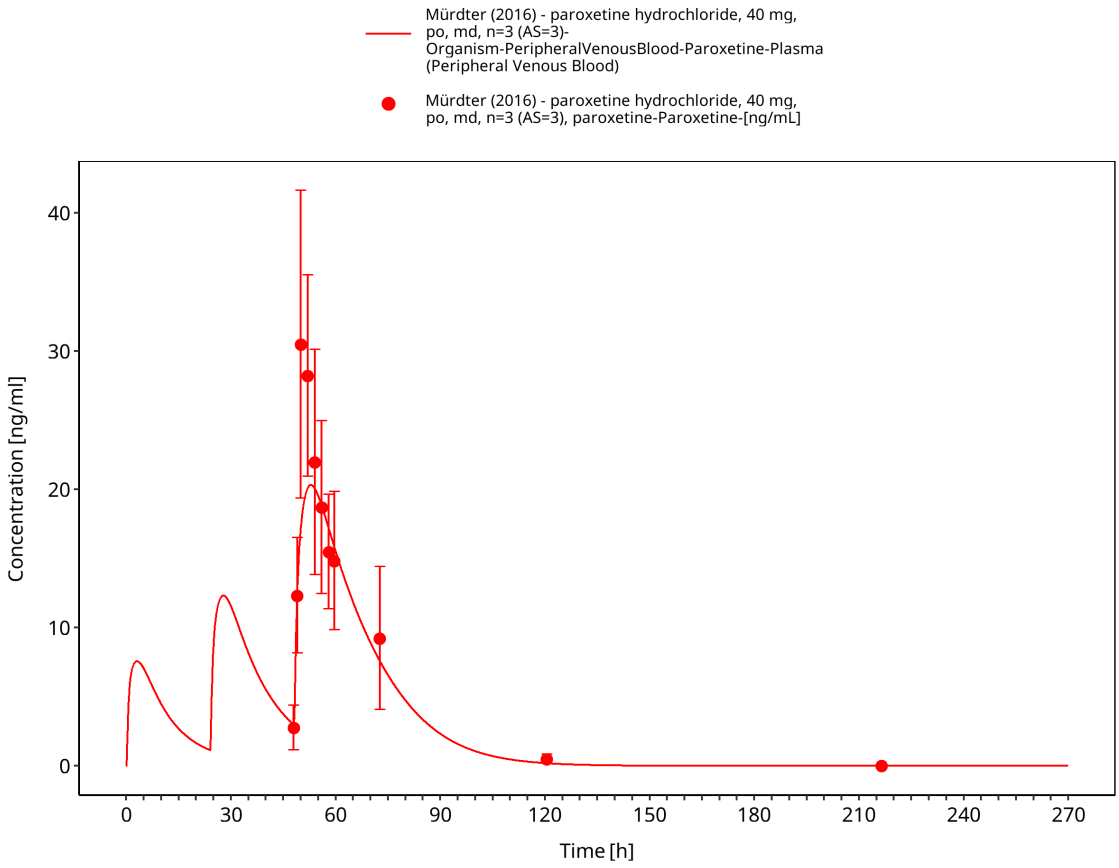


Figure 3-27: Mürdter 2016: paroxetine hydrochloride, 40 mg, po, md, n=3 (AS=3)

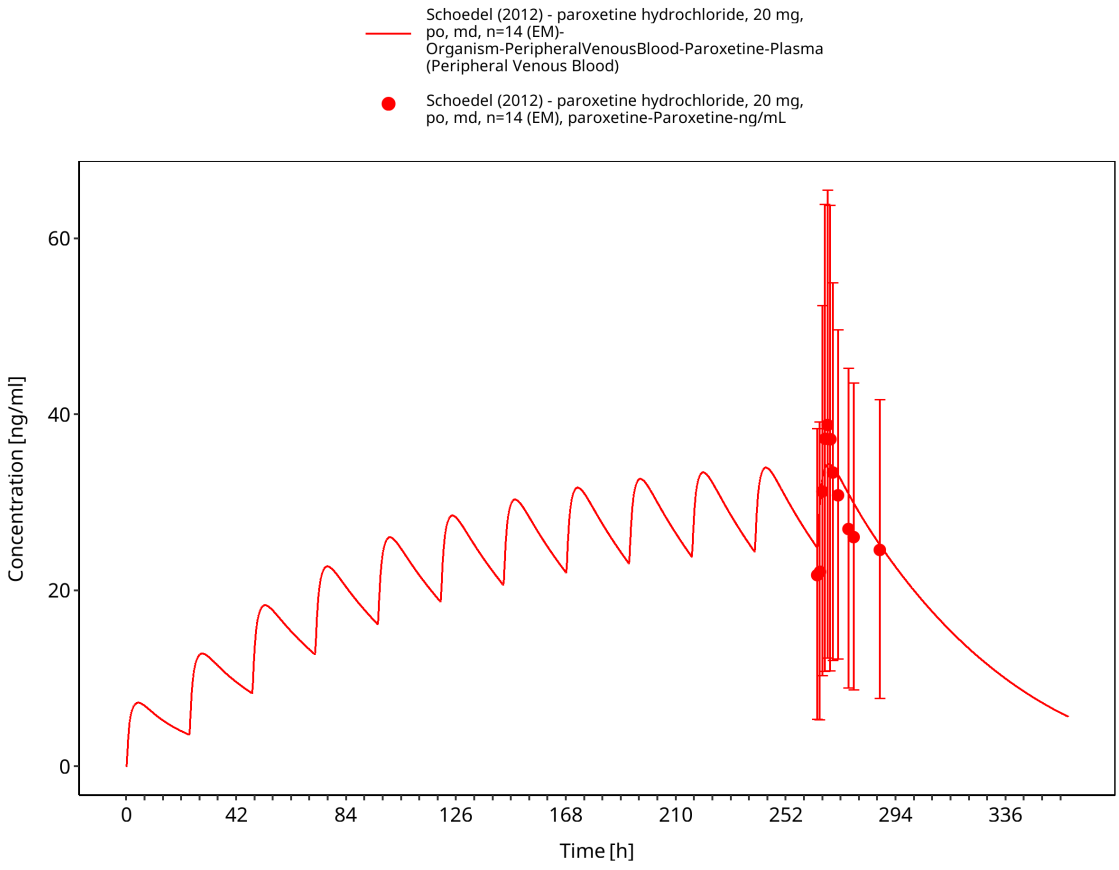


Figure 3-28: Schoedel 2012: paroxetine hydrochloride, 20 mg, po, md, n=14 (EM)

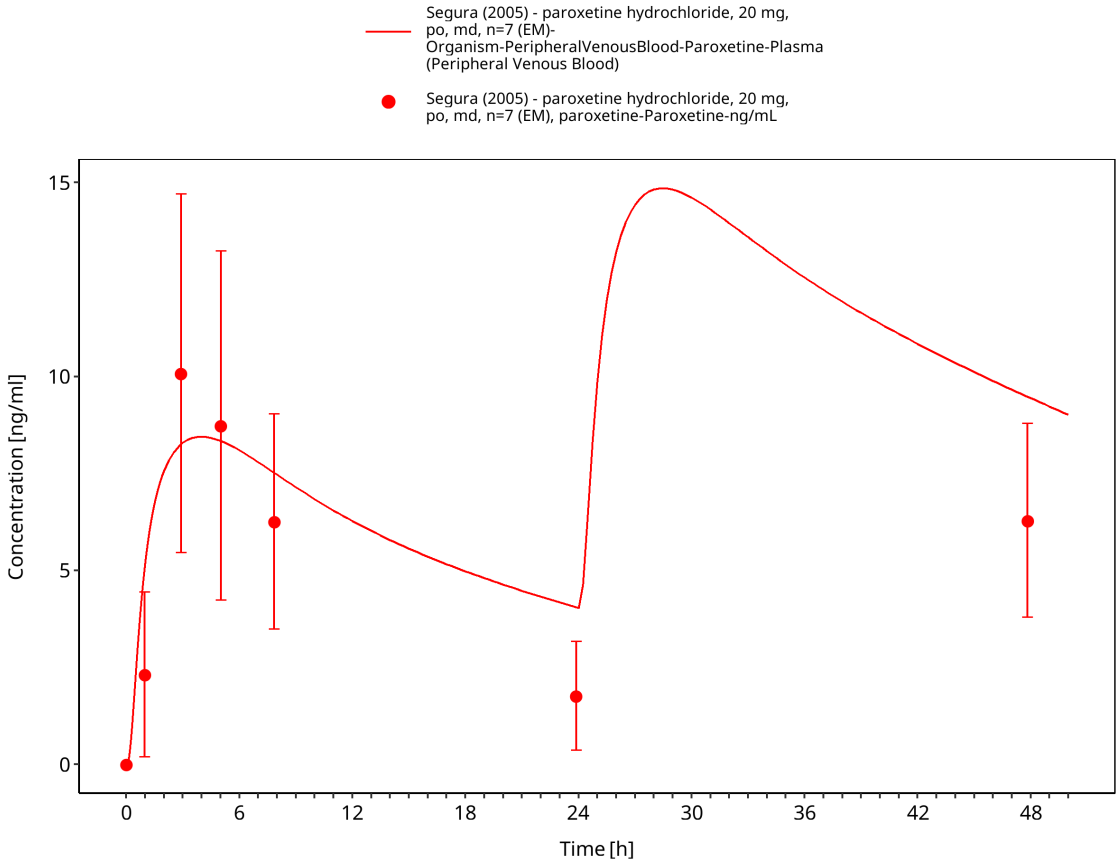


Figure 3-29: Segura 2005: paroxetine hydrochloride, 20 mg, po, md, n=7 (EM)

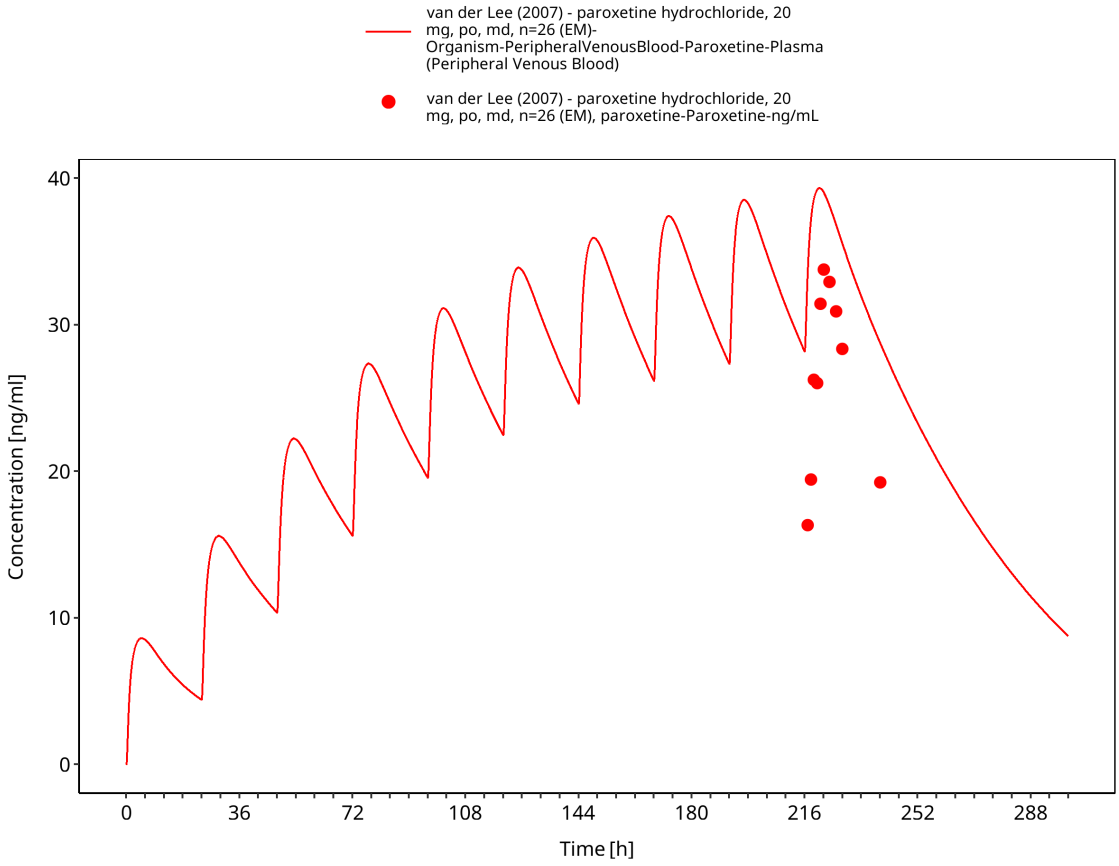


Figure 3-30: van der Lee 2007: paroxetine hydrochloride, 20 mg, po, md, n=26 (EM)

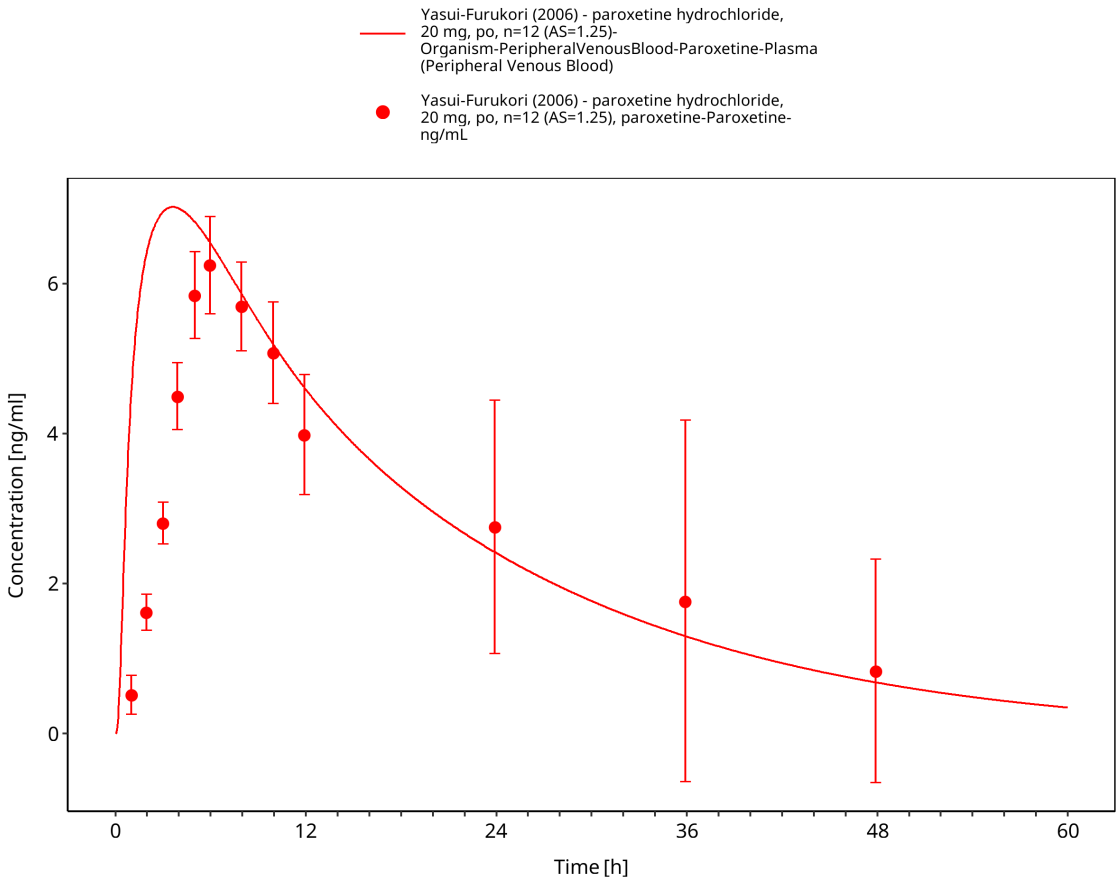


Figure 3-31: Yasui-Furukori 2006: paroxetine hydrochloride, 20 mg, po, n=12 (AS=1.25)

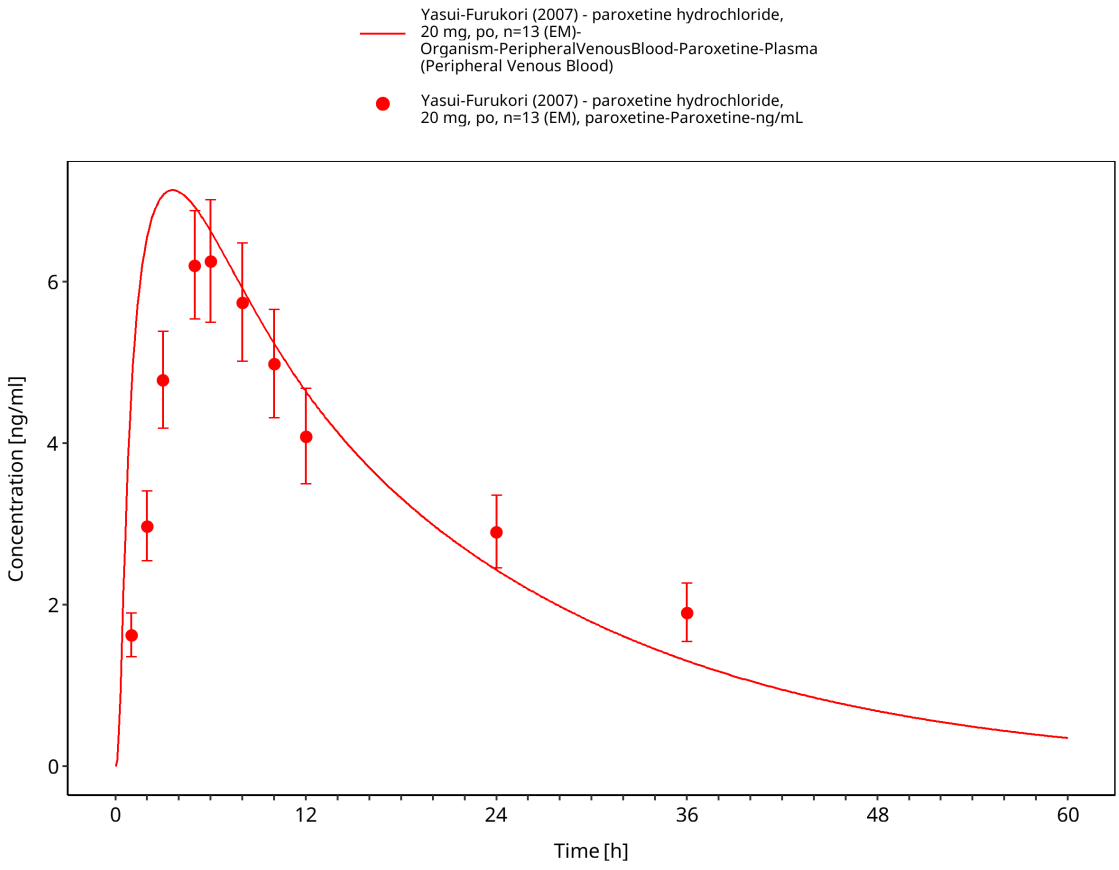


Figure 3-32: Yasui-Furukori 2007: paroxetine hydrochloride, 20 mg, po, n=13 (EM)

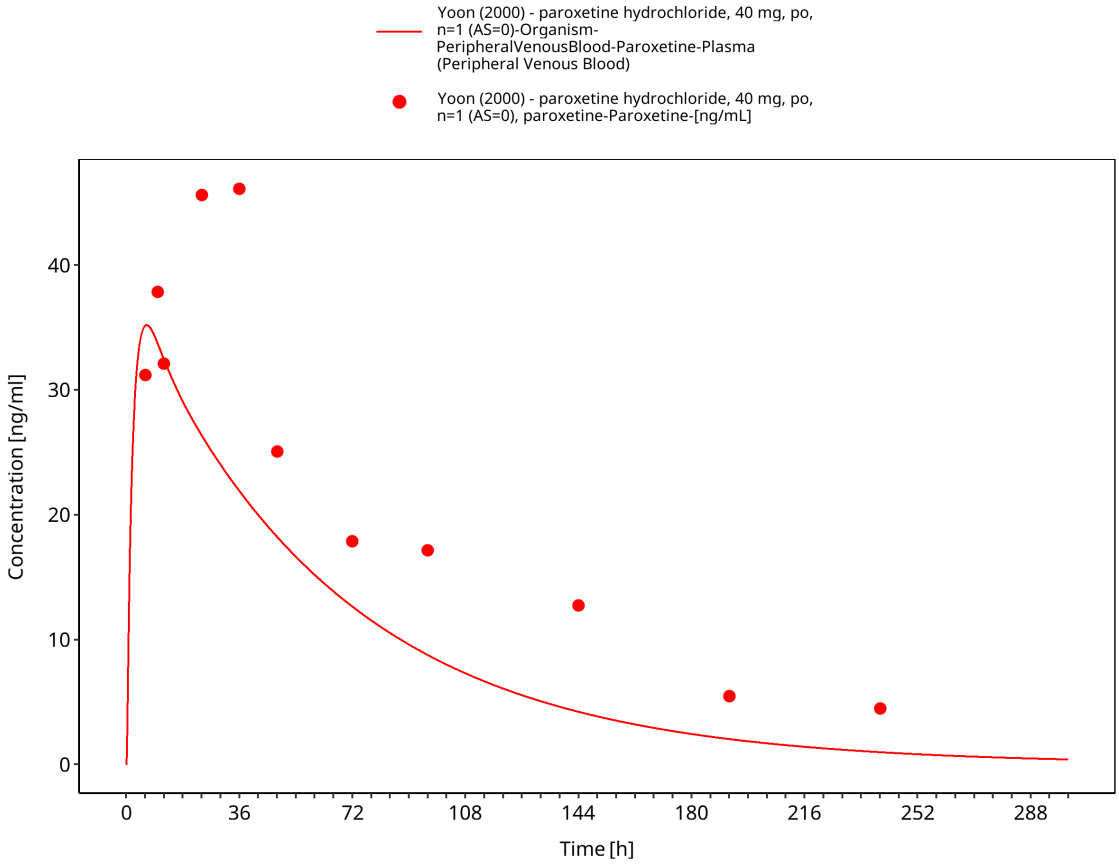


Figure 3-33: Yoon 2000: paroxetine hydrochloride, 40 mg, po, n=1 (AS=0)

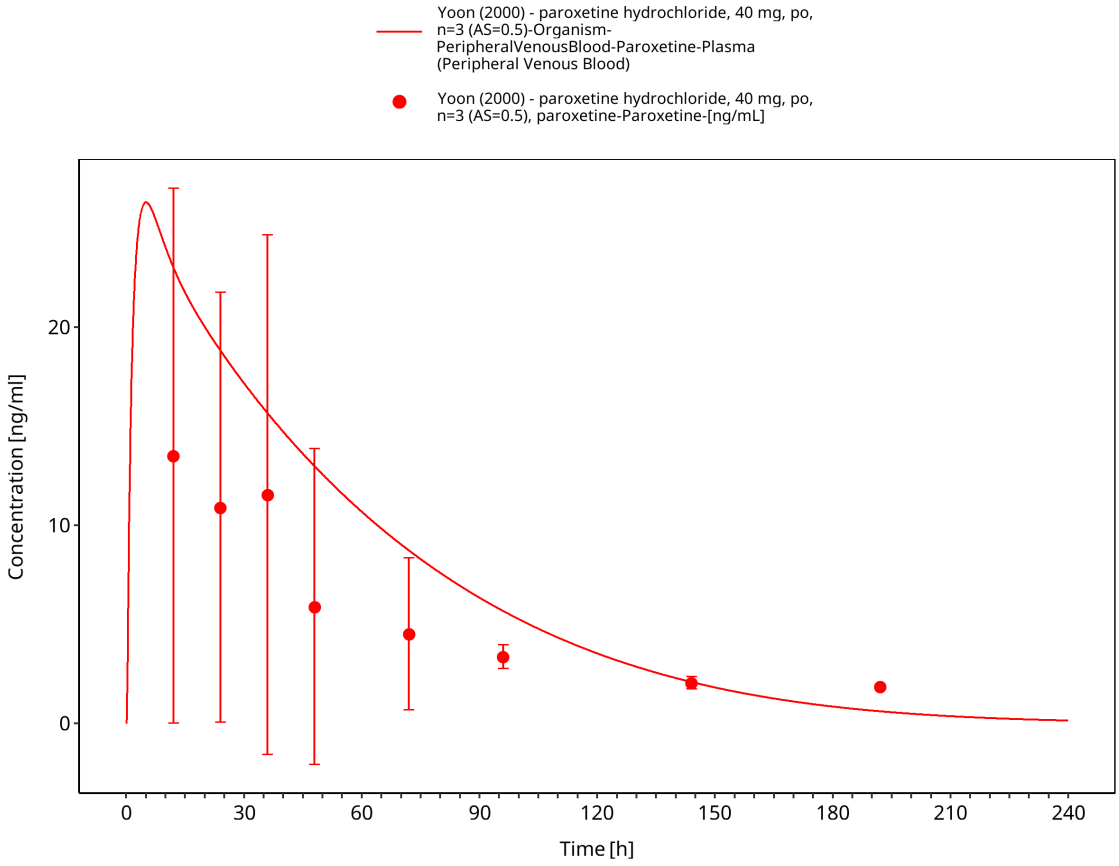


Figure 3-34: Yoon 2000: paroxetine hydrochloride, 40 mg, po, n=3 (AS=0.5)

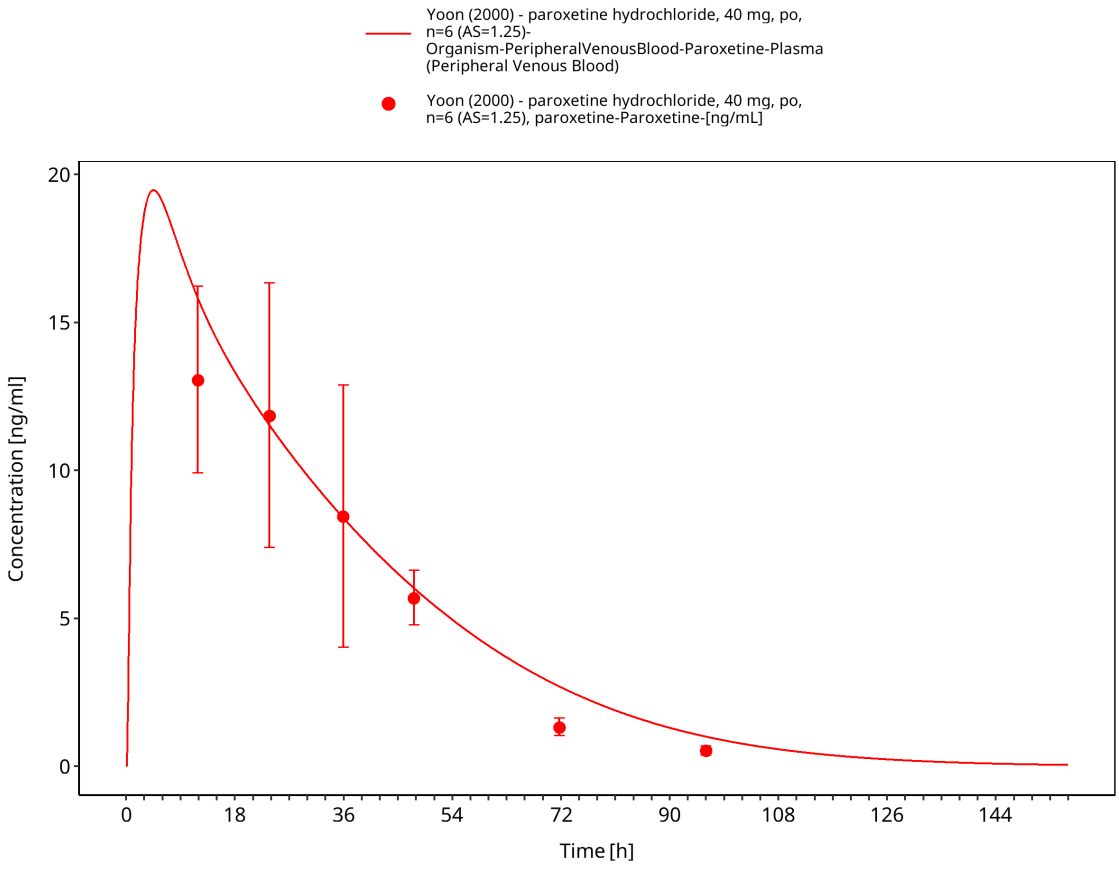


Figure 3-35: Yoon 2000: paroxetine hydrochloride, 40 mg, po, n=6 (AS=1.25)

4 Conclusion

The paroxetine PBPK model describes the evaluated plasma concentration-time data after intravenous doses of 23–28 mg and oral doses of 20–70 mg in adults. The evaluation includes single-dose and multiple-dose administration, immediate-release and controlled-release formulations, and CYP2D6 poor, intermediate, extensive, normal, and ultrarapid metabolizer groups represented by phenotype or activity score.

The model uses common absorption and distribution parameters across CYP2D6 groups. CYP2D6 provides the principal metabolic pathway, while CYP3A4, residual hepatic clearance, and passive glomerular filtration provide additional elimination pathways. Activity-score-specific CYP2D6 catalytic rates are used to simulate exposure differences across the evaluated CYP2D6 groups.

The concentration-time profiles and goodness-of-fit diagnostics characterize model performance within the evaluated dose, route, formulation, regimen, and CYP2D6 ranges. No formal acceptance criterion was applied. This compound report evaluates only paroxetine-only or control study arms.

The main limitations are the lumped representation of hepatic clearance and the absence of explicit metabolites and P-gp transport.

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